Active Geometric Maintenance of the Spinal S-Curve Against Gravity:
An Information–Mechanical Coupling Model for Adolescent Idiopathic Scoliosis Onset

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Abstract

Purpose: Adolescent idiopathic scoliosis (AIS) clusters at peak height velocity, in the thoracic spine, and overwhelmingly in girls (8:1), yet no single framework unifies its timing, location, and sex predominance. We test whether AIS can be modelled as a failure to recover spinal geometry during growth — when growth-driven remodeling outpaces active geometric restoration.

Methods: We model the developing spine as a biomechanical control system: a Cosserat rod whose segment-wise stiffness is set by developmental HOX-domain patterning and stabilised against gravity by proprioceptive feedback. Four research strands: (i) cross-species allometric analysis across a 12-row vertebrate panel (11 species, human at two developmental stages; 10 rows analysed after two pre-specified exclusions) using $\mathcal{B}_g = EI/MgL^2$; (ii) AlphaFold-derived structural metrics (elongation, hinge-density per 100 residues; length-matched controls) for 61 candidate proteins; (iii) a two-timescale recovery-ratchet model of curve setting through the growth spurt (geometric-restoration rate k_r versus growth-remodeling rate k_g); (iv) calibration against a simulated cohort ($N = 1,000$) and bracing simulation compared against the Lonstein & Carlson progression index and the BrAIST clinical trial.

Results: Cross-species, $\mathcal{B}_g$ declines monotonically with body mass (exponent -0.282, SE 0.058, $R^2 = 0.744$), but we report as a negative result that it does *not* isolate the human condition: the human adult ($\mathcal{B}_g = 0.0101$) is indistinguishable from the rabbit and sits above cat, dog, horse and elephant, so the case for obligate active maintenance rests on posture and axial load path rather than on a $\mathcal{B}_g$ threshold. A curated subset of extended-filament/channel mechanosensors (Vimentin, Lamin A/C, PIEZO2, SCRIB, PTK7) is 72% more structurally elongated than metabolic regulators ($p = 0.021$ two-sided, $p = 0.011$ one-sided, Cohen's $d = 1.13$, $n = 23$; signal dilutes with broader gene sets) and has 70% fewer hinge regions per residue ($p = 0.003$, $n = 61$) — consistent with cooperative load-sensing. In the recovery-ratchet simulation, the growth-remodeling rate peaks at peak height velocity and overtakes the geometric-restoration rate, so transient curvature is progressively cemented into permanent set — predicting that curve flexibility (reducibility) declines as the curve progresses; over the peak-height-velocity interval the model-derived demand-to-supply ratio rises by $\sim$29%, a prefactor-free relative-growth prediction rather than a calibrated threshold. The effective Bio-Gravitational parameter B_g^{eff} tracks the Lonstein & Carlson Progression Factor across the simulated cohort ($r = -0.483$); both quantities are deterministic functions of the same four simulated demographic variables, so this is a structural consistency check rather than independent validation, and the nominal p-value is not interpretable as evidence. Simulated bracing (modeled as 37% gravitational load-sharing) reproduces the BrAIST success rates (72% braced vs 48% observation), which were themselves the grid-search targets used to calibrate the three free parameters; the Risser-stratified subgroup rates were *not* calibration targets and replicate in direction but not magnitude (model 70.3%/74.4% vs trial 64%/78%). In a deterministic sweep, the critical spine length L_{crit} *declines* with structural anisotropy $\mathcal{A}$ ($r = -0.88$), identifying reduction of $\mathcal{A}$ as the therapeutic direction; 20 of 50 sweep points lie on the search bounds, so this is reported as a direction rather than a calibrated magnitude.

Conclusions: The framework reframes AIS as a hypothesised failure of an active biomechanical control system, integrating developmental patterning, proprioceptive feedback, and tissue mechanics. The effective Bio-Gravitational parameter is a candidate proxy for progression and brace response; its agreement with the Lonstein–Carlson index is a structural consistency check and its agreement with BrAIST rates is a calibration, so neither constitutes independent validation. Three hypothesis-generating predictions follow, presented *not* as clinical recommendations: (i) curve flexibility (reducibility on supine and bending films) should decline through the growth spurt in adolescents who progress, and low reducibility at presentation should predict progression independent of skeletal maturity; (ii) the model predicts that mitochondrial NAD^+ availability modulates simulated curve progression — **this does not endorse nutraceutical use outside an IRB-approved trial**; (iii) ciliary length and cellular dithering should increase in scoliotic osteoblasts before macroscopic onset.

Impact Statement: We identify a quantitative scaling law governing the metabolic cost of actively maintaining spinal geometry against gravity, explaining why the human spine is uniquely vulnerable to scoliosis during adolescent growth and reframing AIS as a hypothesised failure of an active gravity-loaded postural-control system rather than a stochastic genetic defect.

Model Summary: The Allometric Trap

The Question: What physical law governs how organisms maintain shape against gravity, and why does this maintenance fail during adolescent growth?

The Scaling Law: The Bio-Gravitational Number $\mathcal{B}_g = EI/(MgL^2)$ predicts that humans ($\mathcal{B}_g \approx 0.01$) cannot passively resist gravity and must actively maintain spinal geometry at metabolic cost.

The Mechanism:

- **Metabolic Scaling Mismatch**: The cost of anti-gravitational form maintenance scales as L^3 (elastic strain-energy route), while paraspinal metabolic supply is surface-limited ($\sim L^2$), widening the Energy Deficit Window fastest during peak height velocity.
- **Molecular Asymmetry (Exploratory)**: Initial AlphaFold analysis suggests mechanosensory proteins may exhibit higher structural anisotropy than their metabolic regulators. We present this as a hypothesis-generating observation for future study, acknowledging current AI limitations in predicting mechanical properties.
- **Scoliosis as Scaling Failure**: During adolescent growth, this deficit forces collapse from the active S-curve into a lower-energy scoliotic configuration — a hypothesised failure of the active control system, not a stochastic genetic defect.

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1 Introduction

1.1 The Clinical Problem: Adolescent Idiopathic Scoliosis

Adolescent Idiopathic Scoliosis (AIS) affects 2–4% of adolescents worldwide, representing the most common spinal deformity requiring clinical intervention [1, 2]. Despite over a century of research, the fundamental question remains unanswered: *why does scoliosis initiate during the growth spurt at ages 11–15, and why is the thoracic spine (T8–T10) most vulnerable?*

Current explanatory frameworks — neuromuscular asymmetry [3], genetic predisposition [4], and Hueter-Volkmann asymmetric loading [5] — describe *correlations* but, in isolation, offer no unified mechanism connecting age window, anatomic localization, and 8:1 female predominance. Our framework retains the Hueter–Volkmann law not as a standalone correlate but as the *cementing* step of a quantitative recovery-ratchet mechanism (Theory), the process by which an un-recovered deviation becomes permanent. As a result, treatment remains empirical: observation for mild curves (Cobb angle $< 25°$), bracing for moderate curves (25–45°), and corrective fusion surgery for severe deformity ($> 45°$) [6].

1.2 An Allometric Vulnerability: Why the Human Spine Is Uniquely Exposed

We propose a paradigm shift: AIS is not a random genetic error but a *hypothesised scaling failure* of an active gravity-loaded control system. The human spine sustains an intricate S-shaped profile (cervical/lumbar lordosis, thoracic kyphosis) rather than passively sagging into the C-curve predicted by classical beam mechanics [7, 8]. Maintaining this upright posture against gravity is intrinsically metabolically expensive, requiring continuous mechanosensory feedback via proteins like PIEZO1/2 and primary cilia [9–11].

Maintaining an upright posture in a gravitational field is intrinsically expensive, requiring continuous muscular actuation, proprioceptive feedback, and cellular mechanotransduction to oppose gravity and prevent structural collapse [12–15]. Mechanosensory structures, including primary cilia and highly anisotropic demand-side proteins like PIEZO1/2 and Vimentin, act as critical load sensors, continuously monitoring mechanical strain and mediating metabolic responses [9, 10, 16]. Disruptions in this mechanosensory loop, whether through microgravity unloading or genetic mutations in ciliary or mechanotransductive pathways, lead to profound structural abnormalities such as scoliosis and bone density loss [11, 17, 18]. Consequently, spinal morphogenesis and geometry maintenance cannot be viewed solely as a solid mechanics problem, but as a continuous mechanobiological computation.

1.3 The Bridge: Counter-Curvature as an Active Control System

To understand how the spine maintains its shape and why it is uniquely vulnerable during growth, we model the spine as an active control system that continuously expends metabolic energy to oppose gravity. We refer to this active maintenance as "biological counter-curvature" to distinguish it from passive structural support. Like other dissipative biological control systems [19, 20], the spine sustains its geometry only as long as the metabolic budget supports the proprioceptive feedback loop.

This active control system provides a physical basis for its developmental vulnerability. We formalize this through the **Bio-Gravitational Number**, $\mathcal{B}_g = EI/(MgL^2)$, a dimensionless ratio analogous to the Froude number. Cross-species analysis reveals that most vertebrates maintain $\mathcal{B}_g > 0.1$ (passively stable), but humans occupy a unique "Allometric Trap" ($\mathcal{B}_g \approx 0.01$) where passive stiffness is entirely insufficient and active metabolic maintenance is required [21, 22]. This fragile state requires constant mechanosensory monitoring.

During the adolescent growth spurt, this control system faces a *recovery* crisis. Growth continually perturbs spinal geometry, and the spine must restore its reference shape faster than growth cements the deviation. Restoration competes with growth-driven remodeling: by the Hueter–Volkmann law, a sustained asymmetric deviation is progressively cemented by asymmetric growth-plate activity [5, 23]. When the growth-remodeling rate peaks at peak height velocity (PHV) and overtakes the geometric-restoration rate, transient curvature ratchets into permanent set — a loaded spring that cannot recoil. We propose this **recovery ratchet**, rather than a postural-balance failure, as the onset mechanism (Theory). It explains why onset clusters at PHV, why affected adolescents are otherwise neurologically normal (the fast balance loop is intact), and why curves are detected incidentally; and it predicts a measurable structural signature — declining curve flexibility/reducibility as the curve sets.

Furthermore, this delay instability is exacerbated by an **Energy Deficit Window**. The strain-energy cost of maintaining the active control against gravity scales as L^3, while nutrient supply to the avascular intervertebral disc scales only as L^2 (surface-limited perfusion) [24]. We propose that this transient energy deficit, potentially driving localized NAD+ insufficiency and axonal degradation [25, 26], lowers the metabolically expensive geometric-restoration rate k_r at the very window where the growth-remodeling rate k_g peaks. The coincidence of a depressed restoration rate and a peak remodeling rate during PHV forms the biophysical foundation of scoliosis onset.

1.4 Predictions

By modeling the spine as a two-timescale recovery ratchet (geometric restoration versus growth-remodeling), this model moves beyond correlative observations to a candidate causal mechanism, generating three specific, falsifiable theoretical predictions for hypothesis-generation. We emphasize that these predictions arise from in-silico modeling and **do not constitute clinical recommendations**; prospective testing in IRB-approved cohorts is required before any intervention is suggested:

- **Theoretical metabolic rescue:** The model predicts that mitochondrial cofactor availability (e.g., NAD^+) modulates curve progression in simulation. This generates a hypothesis testable in IRB-approved adolescent cohorts; we do not endorse off-label supplementation in adolescents outside such trials.

- **Declining curve flexibility as a biomarker:** Curve flexibility (reducibility quantified by supine-versus-standing Cobb change, or by bending/traction films) is predicted to decline as κ_p accumulates through the growth spurt in adolescents who progress; low reducibility at presentation should predict subsequent progression independent of skeletal maturity. This is testable in routinely acquired standing, supine, and bending radiographs without new instrumentation.

- **Ciliary length and cellular dithering:** If structural sensing requires active metabolic supply, the model predicts cells under metabolic constraint will upregulate ciliary length or rely on stochastic micro-tremors to maintain signal-to-noise ratios. These are testable in scoliotic osteoblast samples obtained from existing intraoperative tissue banks.

2 Theoretical Framework

2.1 The Bio-Gravitational Number: A Stability Threshold

The central question is: under what physical conditions can an organism maintain its shape against gravity? We derive a dimensionless parameter, the **Bio-Gravitational Number**:

$$\mathcal{B}_g = \frac{EI}{MgL^2}, \tag{1}$$

where EI is spinal flexural stiffness, M is body mass, g is gravitational acceleration, and L is spine length. This ratio compares passive structural resistance to gravitational bending load.

Cross-species analysis does not reveal a threshold: $\mathcal{B}_g$ declines monotonically with body mass (exponent -0.282, SE 0.058, $R^2 = 0.744$; $n = 10$ rows after two pre-specified exclusions), and the human adult ($\mathcal{B}_g = 0.0101$) is indistinguishable from the rabbit and sits above cat, dog, horse and elephant. The case that the normal spinal S-curve (cervical lordosis, thoracic kyphosis, lumbar lordosis) requires continuous metabolic maintenance therefore rests on posture and the axial load path of an upright, single-column stance — not on a uniquely low $\mathcal{B}_g$ — and that maintenance is implemented through cellular strain sensors (PIEZO1/2, primary cilia) and proprioceptive feedback.

2.2 The Energy Deficit Window

During adolescent growth, two unfavorable scaling relationships collide. The metabolic cost of maintaining the S-curve against gravity scales approximately as L^3 (elastic strain energy $U \sim EI\kappa^2 L$ with $I \sim L^4$, $\kappa \sim L^{-1}$). Nutrient supply to avascular intervertebral discs, governed by diffusion and convective transport through endplate surface area, scales as L^2 (surface-area limited) [24]. This widening deficit requires that paraspinal metabolic supply be **surface- (perfusion-) limited**: were supply volume-limited ($\sim L^3$), demand and supply would co-scale and no deficit window would open.

The deficit ratio $R(L) = C\,L^{n-2}$ with $n = 3$ increases monotonically with length; its absolute crossover depends on the composite prefactor C (muscle efficiency, EI, perfusion density), which we do not calibrate here. The **prefactor-free** prediction is relative deficit growth across the growth spurt: $R(0.45)/R(0.35) = (0.45/0.35)^1 \approx 1.29$, i.e. a ∼29% rise in the demand-to-supply ratio over the peak-height-velocity interval. We therefore present the growth-spurt window as the interval of fastest deficit *increase*, not as a single calibrated crossover length. This age-length correspondence matches the well-known clinical peak incidence of Adolescent Idiopathic Scoliosis without parameter fitting to clinical data.

2.3 The Recovery Ratchet: Restoration versus Growth-Remodeling

We interpret AIS onset not as a postural-balance failure but as a *geometry-recovery* failure. Growth continually perturbs the spine from its reference shape; the active system must restore that shape faster than growth cements the deviation. We split the standing curvature deviation into a recoverable elastic component κ_e (the part that reduces supine or on bending films) and a permanently remodeled component κ_p. Two rates compete: the active geometric-restoration rate k_r and the growth-remodeling rate $k_g(t)$, by which a sustained deviation is cemented through Hueter–Volkmann growth-plate modulation [5, 23]. For a perturbation that recoils at rate k_r and is cemented at rate k_g, the fraction converted to permanent set is $k_g/(k_r + k_g)$, giving

$$\frac{d\kappa_p}{dt} = k_g(t)\,\kappa_e\,\frac{k_g(t)}{k_r + k_g(t)}, \qquad \text{flexibility}(t) = \frac{\kappa_e}{\kappa_e + \kappa_p(t)}. \tag{2}$$

When $k_r \gg k_g$ the spine recoils and little sets (flexibility $\to 1$); when $k_g \gtrsim k_r$—as occurs when $k_g(t)$ peaks at peak height velocity (PHV)—each perturbation is partly cemented before it can be recovered, and the deviation *ratchets* monotonically into a permanent curve (Fig. 11). Because $k_g(t)$ is maximal during the growth spurt, the ratchet predicts the clustering of AIS onset at PHV *without* invoking a postural-balance deficit, consistent with affected adolescents being otherwise neurologically normal and curves being detected incidentally. The Energy Deficit Window acts within this framework by lowering k_r—starving the metabolically expensive restoration machinery—and known genetic risk factors (reduced collagen-dependent damping, altered mechanosensation) likewise shift an individual toward the ratchet regime. The model yields a falsifiable structural signature: *curve flexibility (reducibility) declines as κ_p accumulates*, so progressing curves should become progressively less reducible on supine and bending films—a prediction already corroborated by the established clinical association between low curve flexibility and progression [27–29].

2.4 Information-Elasticity Coupling: HOX Genes to Geometry

The model formalizes how developmental patterning genes (particularly HOX genes encoding vertebral identity along the cranio-caudal axis) translate into mechanical geometry. We introduce an **Information Field** $I(s)$ representing the spatial concentration of HOX-encoded positional information along the spine (coordinate s). This field couples to spine mechanics through three channels (see Supplementary Theory for full derivation):

1. **Rest curvature modulation**: Information gradients specify target curvatures (lordotic at cervical/lumbar, kyphotic at thoracic).
2. **Stiffness modulation**: Information modulates local bending and torsional stiffness via extracellular matrix composition.
3. **Active moment generation**: Information gradients drive active muscular moments maintaining the S-curve.

When the energy deficit exceeds a critical threshold, this active control system can no longer suppress small left-right asymmetries (inevitable developmental noise, typically $\sim$ 1–5%). The metabolically expensive feedback loop operates near its delay budget at this stage: at the single operating point of Section 3.6 ($K_P = 120$, $K_D = 12$, $\tau_0 = 45\,\text{ms}$; PHV excursion $K_D \approx 7$, $\tau \approx 75\,\text{ms}$), the loop approaches but does not cross the exact Hopf boundary $\tau_c = 79\,\text{ms}$ (Supplementary Section S10.2) — degraded feedback fidelity acts on restoration quality, not on crossing a stability threshold. Under the recovery ratchet above, such asymmetries—benign during childhood because they are readily recovered—are instead cemented rather than recovered once $k_g \gtrsim k_r$, ratcheting into macroscopic scoliotic deformities (Cobb angles $> 15°$) during the deficit window.

2.5 Why Lateral (Coronal Plane) Buckling?

The spine is loaded primarily in the sagittal plane (forward-backward) by gravity. Why does deformity emerge laterally? The model predicts a **directional-stiffness anisotropy**: developmental patterning gradients (encoding the S-curve) are aligned with gravitational stress in the sagittal plane, providing maximal effective stiffness in that direction. Small lateral asymmetries create orthogonality between information and stress vectors, reducing effective lateral stiffness (31.1% at the profile peak in the reference simulation; the model ceiling as $\alpha \to 0$ is set by the transverse-to-longitudinal modulus ratio, $\sim$80%). This creates a localized "soft spot" in the mid-to-lower thoracic region (T8–T10, the usual apex of right-thoracic AIS, corresponding to the steepest information gradient), directing instability into the coronal plane and explaining the anatomic localization of right thoracic AIS.

Full mathematical derivations, stability analysis, and parameter sensitivity studies are provided in Supplementary Theory Sections S1–S4.

3 Methods

3.1 Computational Framework

We implemented the Information-Elasticity Coupling framework using two complementary approaches: (1) a fast deterministic beam model for parameter space exploration, and (2) full 3D Cosserat-rod simulations (a standard slender-continuum model in spine biomechanics) capturing both bending and twisting under large deformations.

For 3D simulations, we used `PyElastica` [30, 31], an open-source Python implementation of Cosserat rod theory. The spine was modeled as a slender elastic rod ($n = 50$–100 elements) with length $L = 0.3$–$0.5\,\mathrm{m}$, radius $r = 0.01\,\mathrm{m}$, Young's modulus $E_0 = 1\,\mathrm{GPa}$, and density $\rho = 1100\,\mathrm{kg/m^3}$ (representative of vertebra-disc composite properties). Boundary conditions: clamped at base (sacrum), free at top (cranium), under uniform gravitational loading $\mathbf{g} = -9.81\,\mathrm{m/s^2}$.

As an optional acceleration layer, we implemented a GPU-native NVIDIA Warp screening workflow for large Euler–Bernoulli parameter sweeps. This workflow evaluates analytic passive-sag and $\mathcal{B}_g$ quantities across candidate length, radius, stiffness, and density combinations, then forwards selected high-risk regimes to the validated PyElastica beam setup for nonlinear refinement. Warp results are used only for computational triage and reproducibility auditing; all mechanistic claims in the main text remain based on the deterministic beam model, PyElastica Cosserat simulations, and model-internal statistical analyses described below.

3.2 Information Field Specification

The developmental information field $I(s)$ encoding HOX-patterned positional identity was parameterized as a bimodal Gaussian:

$$I(s) = A_c \exp\left[-\frac{(s/L - s_c)^2}{2\sigma_c^2}\right] + A_l \exp\left[-\frac{(s/L - s_l)^2}{2\sigma_l^2}\right] + I_0, \tag{3}$$

with peaks at normalized positions $s_c = 0.80$ (cervical) and $s_l = 0.25$ (lumbar), widths $\sigma_c = 0.08$ and $\sigma_l = 0.10$, amplitudes $A_c = 0.5$ and $A_l = 0.7$, and baseline $I_0 = 0.3$. This functional form reproduces the anatomic distribution of lordotic (high I) and kyphotic (low I) regions.

For scoliosis simulations, a small lateral asymmetry was introduced: $I(s) \to I(s) + \varepsilon_{\text{asym}} \exp[-(s/L - 0.6)^2/(2 \cdot 0.08^2)]$ with $\varepsilon_{\text{asym}} = 0.01$–$0.05$ (1–5% left-right variation, representing normal developmental noise).

3.3 Information-Elasticity Coupling Implementation

The information field couples to spine mechanics through three channels: (1) rest curvature modulation: $\boldsymbol{\kappa}^0(s) = \boldsymbol{\kappa}_{\text{gen}} + \chi_\kappa \nabla I(s)$, (2) stiffness modulation: $B(s) = E_0 I_{\text{area}}(1 + \chi_E I(s))$, and (3) active moment generation: $\mathbf{M}_{\text{active}}(s) = \chi_M I'(s)\hat{\mathbf{n}}(s)$. Coupling constants χ_κ, χ_E, χ_M were systematically varied to explore the parameter space. We implemented custom callbacks in PyElastica to update local material properties and rest configurations based on $I(s)$ at each timestep.

3.4 Bio-Gravitational Number Calculation

For cross-species analysis, we computed $\mathcal{B}_g = EI/(MgL^2)$ using published allometric data for a 12-row panel (11 species spanning mouse to dolphin, with Human-Adult and Human-Child as separate developmental stages of the same hominin species; giraffe and dolphin are pre-specified exclusions, so all statistics use 10 rows; we acknowledge this limits independent biped sampling and recommend extension to gibbon, orangutan, and bonobo for stronger phylogenetic claims). Spine length L, body mass M, and vertebral dimensions were obtained from comparative anatomy databases. Flexural stiffness EI was estimated from vertebral cross-sectional geometry and published elastic moduli for bone-disc composites. Log-log regression was performed to extract scaling exponents with 95% confidence intervals via bootstrap resampling (10,000 iterations).

3.5 Energy Deficit Window Quantification

Metabolic demand for counter-curvature maintenance was modeled as $D(L) \propto L^3$ (elastic strain-energy route: $U \sim EI\kappa^2 L$). Nutrient supply capacity to avascular discs was modeled as $S(L) \propto$

L^2 (diffusion-limited by endplate surface area) [24]. The relative deficit growth $R(L)/R(L_0) = (L/L_0)$ was mapped to the peak-height-velocity interval using WHO pediatric spine growth charts.

3.6 Postural Feedback Operating Point

Where the delayed proprioceptive feedback loop enters the analysis (lumped model and characteristic equation, Supplementary Section S10.2), a single operating point is used throughout this work: $K_P = 120$, $K_D = 12$, resting feedback delay $\tau_0 = 45\,\text{ms}$, with the growth-spurt (peak-height-velocity) excursion to $K_D \approx 7$ at $\tau \approx 75\,\text{ms}$. The delay margin of this plant is bounded: the exact Hopf boundary (Supplementary Eq. S18) peaks at $\tau_c = 79\,\text{ms}$ at $K_D^* = 12.4$, so the operating point sits inside the stable region at rest and approaches — but does not cross — the boundary at PHV. Earlier drafts quoted $K_D = 8$ with $\tau_0 = 150\,\text{ms}$; that point lies past the boundary ($\tau_c = 67\,\text{ms}$ at $K_D = 8$) and is retired.

3.7 Outcome Metrics

For each simulation, we computed: (1) **Equilibrium deviation** $\widehat{D}_{\text{geo}}$ quantifying departure from the gravity-only passive equilibrium configuration, (2) **Lateral scoliosis index** $S_{\text{lat}} = \max |y|/L_{\text{eff}}$ measuring coronal-plane asymmetry, (3) **Cobb angle** via linear fits to upper and lower spine segments (standard clinical measure), and (4) **Lordosis/kyphosis angles** from sagittal curvature profiles, compared against clinical norms (cervical lordosis 40–60°, thoracic kyphosis 20–45°, lumbar lordosis 40–60°) [7].

3.8 Clinical Cohort and Bracing Simulation

We calibrated the model's clinical translation using a simulated cohort of $N = 1,000$ pediatric patients (80% female, ages 10–15, initial Cobb angles 10°–30°, Risser stages 0–4). Patient age and sex were mapped to spinal length L and total body mass M using standard pediatric anthropometric growth charts: spinal length was interpolated using Dimeglio growth data, and body mass was interpolated using WHO weight-for-age charts. For prospective registry validation where direct radiographic spine-length measurement is unavailable, L is approximated as $L = 0.31 \times h_{\text{stand}}$ (Dimeglio scaling from standing height h_{stand}); errors-in-variables correction is applied to B_g^{eff} estimates to account for proxy uncertainty.

Skeletal maturity was mapped from Risser sign $R \in [0, 5]$ to spinal bending stiffness $E_0 I(R) = E_0 I_{\text{base}} e^{\alpha R}$, where $E_0 I_{\text{base}} = 1.2\,\text{N}\cdot\text{m}^2$ and $\alpha = 0.25$, modeling the exponential increase in flexural rigidity with progressive vertebral ossification. Cobb angle Φ (in degrees) was converted to radians and coupled to geometric stiffness reduction under gravity: $B_g^{\text{eff}} = B_g/(1 + \mu \Phi_{\text{rad}}^2)$ with a geometric scaling factor $\mu = 1.8$.

We calculated the clinical Lonstein & Carlson Progression Factor $PF = (\Phi - 3R)/\text{Age}$ and its corresponding progression probability $P_{\text{prog,clinical}} = 1/(1+e^{-k(PF-PF_0)})$ with $PF_0 = 1.5$ and $k = 1.7$. The model's progression probability was defined as a transition sigmoid $P_{\text{prog,model}} = 1/(1 + e^{\lambda(B_g^{\text{eff}} - B_{g,c})})$. We ran a grid search to calibrate the model's parameters to match the BrAIST trial's overall outcomes: $B_{g,c} = 0.0180$ and $\lambda = 110.0$. Bracing wear was simulated by introducing a load-sharing factor that reduces gravitational mass by 37% ($M_{\text{eff}} = 0.63M$). Simulated success rates (defined as skeletal maturity without progression to a Cobb angle $\geq 50°$) were computed and compared to the BrAIST trial overall and subgroup outcomes.

3.9 AlphaFold Protein Structural Analysis

We retrieved AlphaFold v6 predicted structures for 23 mechanotransduction and metabolic regulator proteins from the AlphaFold Database. Structural anisotropy (ratio of principal moments of inertia) and disorder fraction (DSSP secondary structure analysis) were computed as proxies for structural maintenance cost. Statistical comparison between demand-side (mechanosensors: PIEZO1/2, Vimentin, Lamin A/C) and supply-side (metabolic regulators: SIRT1, PGC-1α, ARNTL) protein classes was performed via Mann-Whitney U test. This analysis is exploratory; AlphaFold limitations in predicting mechanical properties under load are acknowledged [32].

3.10 Numerical Validation

Convergence was verified by varying element count (n = 10–200); deviation from the gravity-only equilibrium converged to within 1% for $n \geq 50$. The PyElastica implementation was validated against the analytical Euler-Bernoulli cantilever solution in the gravity-only small-deflection limit; relative tip-deflection error was below 10% across n = 20–80 elements and decreased with mesh refinement. The optional NVIDIA Warp screen was benchmarked separately as a high-throughput pre-filter and was not used as an independent validation of clinical behavior. All source code is available in the `spinalmodes` Python package (see Data Availability).

Detailed methods including steered molecular dynamics protocols, tissue homogenization calculations, and protein dynamics modeling are provided in Supplementary Methods Sections S5–S7.

4 Results

4.1 Cross-Species Validation: The Allometric Trap

We computed the Bio-Gravitational Number $\mathcal{B}_g = EI/(MgL^2)$ for a 12-row vertebrate panel (11 species, human at two developmental stages) spanning more than five orders of magnitude in body mass (Table 3). Giraffe and Dolphin were pre-specified exclusions — cervical specialisation and non-gravitational (aquatic) sagittal loading respectively — implemented in `scripts/analysis/cross_species_scaling.py`; all reported statistics are for the remaining $n = 10$ rows. Log-log regression of $\mathcal{B}_g$ against body mass yields an allometric scaling exponent of -0.282 (SE 0.058; $R^2 = 0.744$, $p = 1.31 \times 10^{-3}$; percentile bootstrap 95% CI $[-0.347, -0.117]$, 10,000 resamples). This interval contains the -0.25 exponent of Kleiber's law for mass-specific metabolic rate, but it is wide enough to contain a range of alternative exponents and therefore does not discriminate between competing scaling mechanisms.

The data do *not* support a posture-based dichotomy in $\mathcal{B}_g$, and we report this as a negative result. On the present stiffness estimates only the mouse exceeds 0.1; every larger vertebrate falls below it, quadruped and biped alike. The human adult ($\mathcal{B}_g = 0.0101$) is indistinguishable from the rabbit (0.0100) and sits *above* — that is, is nominally more passively stable than — cat (0.0076), dog (0.0045), horse (0.0036) and elephant (0.0022); the excluded giraffe (0.0032) likewise sits below the human. $\mathcal{B}_g$ alone therefore does not single out the human spine, and the case for obligate active maintenance must rest on posture and gravitational load path — the human spine is loaded as a vertical column carrying axial compression rather than as a horizontal beam in bending — rather than on a $\mathcal{B}_g$ threshold. We retain $\mathcal{B}_g$ as a scaling descriptor, not as a discriminator of bipedality. Direct simulation across the full $\mathcal{B}_g$ range $[0.003, 8000]$ (extended sweep, 1920 simulations covering humans through dolphins; Figure 2) shows the human biped regime is structurally distinct from the quadruped regime in countercurvature efficiency η_{CC} ($\eta_{CC} \approx -0.7$ at biped $\mathcal{B}_g$ vs $\eta_{CC} \approx -42$ at quadruped $\mathcal{B}_g$). **Statistical caveat:** the biped subset comprises two developmental stages of a single hominin species (Human-Adult, Human-Child); the formal MWU $p = 0.044$ on per-species means reports a within-species sweep rather than independent phylogenetic sampling, and this biped-vs-quadruped comparison does not survive false discovery rate correction ($q = 0.077$, Benjamini–Hochberg adjusted; see Section §3.3). The qualitative simulation-curve regime contrast remains the load-bearing observation; extension to additional bipedal or facultatively-bipedal species is identified as a high-priority follow-up. Human-adult $\mathcal{B}_g = 0.0101$ and Human-child $\mathcal{B}_g = 0.0201$ are resolved as distinct points on the simulated curve, though they are separated by less than the spread of the quadruped values and the regime contrast should not be read as posture-specific. We note that across the extended sweep, η_{CC} varies smoothly and monotonically with χ_M within each scale — the extended dataset does not support a sharp critical threshold in χ_M at biped $\mathcal{B}_g$ values. The clinical progression of Cobb angle is therefore most likely a smoothly graded function of the underlying coupling parameters rather than a bifurcation crossing, consistent with the gradual character of curve progression observed in pediatric cohorts.

4.2 The Energy Deficit Window

We quantified the metabolic cost of maintaining the S-shaped counter-curvature across the developmental length range $L \in [0.25, 0.55]$ m. The integrated strain-energy demand for sustaining the active S-curve scales as L^3 (Fig. 10). In contrast, diffusive nutrient supply to the avascular intervertebral disc scales as L^2 (surface-area limited). The demand-to-supply ratio therefore rises monotonically through adolescence; across the growth spurt from $L = 0.35$ m to $L = 0.45$ m the relative deficit increases by $\sim$29% ($R(0.45)/R(0.35) \approx 1.29$), defining the **Energy Deficit Window** where AIS risk peaks during peak height velocity (ages 11–15).

The time course demonstrates peak vulnerability matching the clinical onset window. Across the simulated length range $L \in [0.25, 0.55]$ m the predicted Cobb amplitude rises monotonically as $\mathrm{Cobb}(L) \approx aL + bL^3$. We deliberately report *no* correlation statistic for this relationship: L is a uniformly swept grid rather than a sample, and Cobb is a deterministic function of L, so any monotone model yields $r > 0.96$ and the nominal p-value is a function of grid density rather than evidence. A comprehensive summary of the independently computed model-internal statistical tests supporting the IEC framework is provided in Table 7.

4.3 The Protein Cost Landscape

AlphaFold structural analysis (v6) of the 23-protein thermodynamic panel reveals the molecular basis of the Energy Deficit Window (Table 5, generated directly from `research/alphafold_v6_analysis/prot` by `scripts/analysis/regenerate_table_thermodynamic.py`). Demand-side mechanosensory proteins exhibit higher structural anisotropy than supply-side metabolic regulators:

- **Demand (n=12):** Mean anisotropy 3.08 ± 1.50. The five most anisotropic proteins are all demand-side: Vimentin (5.57), PTK7 (5.28), Lamin A/C (4.71), DSTYK (3.65), and PIEZO2 (3.45).
- **Supply (n=11):** Mean anisotropy 1.79 ± 0.59.
- **Gap:** 1.72
  times mean ratio; $p = 0.021$ two-sided ($p = 0.011$ one-sided, Mann–Whitney U), Cohen's $d = 1.13$. Note: this is an exploratory, hypothesis-generating finding, as AlphaFold currently faces limitations in inferring mechanical properties under load (Gomes 2022, Zonta 2024). The distributions overlap substantially — supply-side PLOD1 (3.13) and GHR (2.27) individually exceed eight of the twelve demand proteins — and the finding is sensitive to panel definition (see panel-expansion sensitivity below).

Extended-cohort sensitivity: re-running the same demand-vs-supply categorization against an expanded panel of 26 categorized proteins from a 61-gene AlphaFold dataset (afcc pipeline, see Methods §3.4) yields a stronger result: anisotropy gap of 80% with $p = 0.0094$ and Cohen's $d = 1.22$ after dropping a single high-anisotropy outlier per group (POC5, demand; CDKN1A, supply). This larger-cohort consistency supports the robustness of the demand-supply structural dichotomy. A complementary structural signature emerges in the same expanded panel: mechanosensors carry $\sim 70\%$ fewer hinge regions per residue than other candidate proteins (mean 0.93 vs 3.12 per 100 residues, fold-difference 0.30, Mann–Whitney $p = 0.003$; see Figure 4). **This finding is fully robust to single-gene removal:** leave-one-out cross-validation across all 61 genes yields LOO median $p = 0.003$, max $p = 0.004$, with 100% of LOO iterations remaining significant at both $p < 0.05$ and $p < 0.01$ thresholds — a markedly more robust signal than the demand-supply anisotropy gap discussed above. The combination of high anisotropy and low internal flexibility is consistent with mechanosensors operating as cooperative rigid-body conformational switches rather than flexible-jointed sensors.

Multiple-comparison correction: across the seven independent statistical tests reported in this section (anisotropy gap n=23, anisotropy gap extended n=26 trimmed, hinge density n=61, AIS-strict-tagged vs other, AIS-strict-tagged trimmed, disorder × anisotropy in mechanosensor subset, biped vs quadruped MWU), Benjamini–Hochberg correction yields adjusted q-values: hinge density $q = 0.018$, anisotropy gap (extended trimmed) $q = 0.033$, anisotropy gap ($n = 23$, scored two-sided) $q = 0.049$ — all surviving the $q < 0.05$ FDR threshold. The biped/quadruped MWU does not survive correction ($q = 0.077$); we accordingly downgrade the simulated phylogenetic claim from quantitative to qualitative regime contrast (already noted in Section §3.1). All other tests remain non-significant after correction (raw and adjusted; Supplementary Table S2). A sensitivity analysis excluding 7 low-confidence proteins (pLDDT < 70, such as POC5, GHR, and MESP2) confirms these trends remain robust. Critically, the structural anisotropy signal is *not* confounded by literature popularity: across the 61 genes with both AFCC metrics and curated clinical-evidence priority scores, no correlation is observed between the curated priority score and anisotropy (Spearman $\rho = 0.016$, $p = 0.90$; top vs bottom quintile MWU $p = 0.71$). The demand-vs-supply structural dichotomy is therefore mechanism-specific to the functional categorization, not a citation-density artifact.

Leave-one-out robustness: dropping each gene one at a time and recomputing the demand-vs-supply MWU yields a median LOO $p = 0.041$ (baseline $p = 0.035$), with 84.6% of LOO iterations remaining significant at $p < 0.05$. The result is fragile to removal of any single supply-side gene (LOO $p = 0.092$ when dropping IGF1R, CDKN1A, SIRT1, or PPARGC1A) — reflecting the small supply-side cohort ($n = 7$).

Demand-side expansion sensitivity: extending the demand panel from the original 19 carefully-curated extended-filament/channel proteins to all 37 mechanotransduction/cilia/cytoskeleton-tagged proteins in the AFCC dataset (adding bulky cytoskeletal proteins such as DMD, MYLK, FLNA, FLNB, TUBB) *dilutes the gap*: relative-gap drops from 93% to 42% and Mann-Whitney p rises from 0.035 to 0.33 (Cohen's d from 0.67 to 0.39). This is informative: the demand-supply structural distinction is therefore best characterized narrowly as **extended-filament/channel mechanosensors** (Vimentin, Lamin A/C, PIEZO2, SCRIB, PTK7, CELSR1, VANGL1/2) versus metabolic regulators — not as a general property of all mechanotransduction-tagged proteins. We acknowledge this narrowing in the Discussion. Expanding the supply-side AFCC panel beyond the current $n = 7$ remains the highest-priority next step. The higher disorder fractions (mean 0.42) in thermogenesis-linked supply proteins (like PPARGC1A, ARNTL, SIRT1, GHR, IGF1R) compared to demand-side sensory (η_p) and actuation (η_a) proteins reflect their role as intrinsically disordered signaling hubs. Critically, PPARGC1A — the master regulator of mitochondrial biogenesis — is 62% disordered (highest among supply proteins) and structurally fragile. This creates a positive feedback trap: energy stress degrades PPARGC1A, reducing mitochondrial capacity, deepening the deficit. The model predicts a hypothetical failure sequence (the "VIM Cascade") proceeding from the most anisotropic proteins downward: Vimentin $\rightarrow$ Lamin A/C $\rightarrow$ PIEZO2, which we hypothesize progressively blinds the spine to geometric errors.

4.4 S-Curve Emergence from First Principles

The IEC beam model selects the spinal S-shape as the stable equilibrium configuration of the coupled information-mechanical system. Eigenmode analysis (Fig. 5) shows that for a passive beam ($\chi_\kappa = 0$), the lowest-energy equilibrium is a monotonic C-shaped sag. As χ_κ increases beyond 0.05, a mode crossing occurs: the S-shaped mode becomes the lowest-energy eigenstate, with eigenvalue reduction of ∼30% relative to the C-mode.

Full 3D Cosserat rod simulations (Fig. 7) with human-like parameters ($E_0 = 1\,\mathrm{GPa}$, $L = 0.4\,\mathrm{m}$) reproduce characteristic spinal angles: predicted lumbar lordosis of ∼42° and thoracic kyphosis of ∼35°, within clinical norms (40–60° and 20–45° respectively [7]). Sensitivity analysis (±10% parameter variation) confirms robust counter-curvature ($\widehat{D}_{\mathrm{geo}} = 0.113 \pm 0.011$). Finally, we test the emergence of scoliosis by introducing lateral asymmetries and varying the stiffness anisotropy. As shown in Figure 9, the system exhibits a *bifurcation* sensitive to both the IEC coupling strength χ_κ and the material anisotropy. Systematic sweeps across the parameter space reveal that for a constant growth drive $\chi_\kappa = 5.0$, the system remains stable with Cobb angles $< 10°$ for anisotropy ratios > 2.0. However, as anisotropy drops below 1.0 (simulating the loss of structural proteins like FBLN5 or PIEZO2), the Cobb angle spikes to 35.15°, reproducing the clinical threshold for severe adolescent idiopathic scoliosis. A second, distinct quantity must not be confused with this stiffness ratio: the *structural* anisotropy $\mathcal{A}$ of the mechanosensory proteome, which enters the model as a multiplier on maintenance *demand* ($P_{\mathrm{eff}} = P_{\mathrm{counter}}\, \mathcal{A}/\bar{\mathcal{A}}$). Sweeping $\mathcal{A}$ over 1.0–8.0, the critical spine length L_{crit} *declines* monotonically ($r = -0.88$, $R^2 = 0.775$): higher structural anisotropy raises maintenance cost, so the supply–demand crossover is reached at a shorter spine. The therapeutic direction is therefore *reduction* of $\mathcal{A}$, which delays crossover. We attach no p-value: $\mathcal{A}$ is a swept grid, not a sample. This sweep is additionally bounded — 20 of 50 points sit on the search limits (16 at $L_{\min} = 0.20\,\mathrm{m}$, 4 at $L_{\max} = 0.80\,\mathrm{m}$) — so the apparent plateau above $\mathcal{A} \approx 6$ is the search floor rather than a saturation effect, and the fitted slope should be read as a direction, not a magnitude. Subject to that caveat, scoliosis appears as a stable alternative configuration that the spine drops into when the active control system loses stiffness and becomes hyper-responsive to growth-pattern asymmetries.

This shape persists under gravitational unloading: $\widehat{D}_{\mathrm{geo}}$ remains stable at ≈ 0.091 even as $g \to 0$ (Fig. 7D), explaining why astronauts maintain spinal S-curves in microgravity despite disc expansion and height increases of up to 5 cm [33].

4.5 Phase Diagram and Instability Regimes

We map the full parameter space (χ_κ, g) (Fig. 8). Three distinct regimes emerge: gravity-dominated ($\widehat{D}_{\mathrm{geo}} \approx 0.059$; passive sag), cooperative ($\widehat{D}_{\mathrm{geo}} \approx 0.150$; stable S-curve), and information-dominated ($\widehat{D}_{\mathrm{geo}} > 0.3$; potential instability). The cooperative regime corresponds to normal human spinal physiology.

4.6 Length-Dependent Loss of Postural Stability

We simulate a pubertal growth spurt by increasing L from 0.35 m to 0.45 m. Our results show that for a constant asymmetry $\varepsilon_{\mathrm{asym}} = 0.01$, the Cobb angle remains $< 5°$ for $L < 0.35$ m but undergoes a supercritical bifurcation as length enters the fastest deficit-growth interval. Beyond this threshold, the metabolic instability ratio $R = P_{\mathrm{counter}}/S_{\mathrm{proprio}}$ exceeds unity, and the system enters the information-dominated regime where strain-energy demand (L^3) outpaces surface-limited nutrient supply (L^2). Across $L = 0.35$–0.45 m the relative deficit rises by ∼29%. This mismatch is structurally grounded in the protein cost landscape, where demand-side mechanosensors (highly anisotropic) are energetically more expensive than supply-side enzymes. Furthermore, mapping the peak-height-velocity interval against standard WHO/CDC pediatric spine growth charts retrospectively confirms an age correspondence of roughly 11–12 years, independently matching the known clinical peak incidence window for AIS.

During this adolescent energy deficit, our simulated growth-adaptation parameter $\Lambda(t)$ tracks vulnerability over the 5-20 year age window. The time course demonstrates a distinct peak vulnerability coinciding with the clinical onset window of ages 11-15. Within the simulation, predicted Cobb-angle severity rises monotonically with modelled spinal length; we report no correlation coefficient, since both quantities are deterministic functions of the same swept grid variable. Bifurcation analysis across 400 parameter points places 78.2% of the sampled growth parameter space in energy deficit, with a maximum deficit ratio of 38.5 — both figures are properties of the chosen sweep bounds and carry no sampling interpretation. These findings suggest that rapid axial elongation outpaces the rate at which the geometric-restoration machinery can be metabolically sustained, leaving the spine energetically insolvent and vulnerable to asymmetric buckling into the scoliotic mode. In this deficit state, the restoration rate k_r of Eq. 2 falls relative to the growth-remodeling rate k_g, driving the ratchet toward progressive cementing of curvature.

Introducing lateral asymmetries ($\varepsilon_{\text{asym}} = 0.01$–$0.05$) reveals a supercritical bifurcation at L_{crit} (Fig. 9). Below L_{crit}, asymmetries are suppressed (Cobb $< 5°$). Above L_{crit}, small patterning errors ($\sim$5%) are amplified into macroscopic deformities (Cobb $> 15°$) — the hallmark of adolescent idiopathic scoliosis.

The directional-stiffness anisotropy mechanism explains why buckling is specifically lateral: the alignment parameter $\alpha(s) = |\hat{n}_{\text{info}} \cdot \hat{n}_{\text{stress}}|$ quantifies misalignment between the developmental-patterning gradient and the gravitational-stress direction. A lateral perturbation drives $\alpha \to 0$, reducing effective stiffness toward the transverse modulus (a $\sim$80% ceiling as $\alpha \to 0$; the computed reduction at the profile peak is 31.1%, below). In healthy spines, $\alpha \approx 0.95$ (information and stress vectors aligned); in scoliotic regions, α drops to $\sim$0.3, creating a localized "soft spot" that directs the instability into the coronal plane.

4.7 Testable Predictions

Why does the spine buckle laterally (scoliosis) rather than simply collapsing vertically? Our vector field analysis identifies **directional-signal loss** as the localizing mechanism. We computed the alignment parameter $\alpha(s) = |\hat{n}_{info} \cdot \hat{n}_{stress}|$ along the spinal axis.

- **Healthy Spine:** The information gradient vector $\hat{n}_{info}$ (encoding the S-curve) is largely parallel to the gravitational stress vector $\hat{n}_{stress}$, maintaining high alignment ($\alpha \approx 0.95$) and maximal effective stiffness ($E_{eff} \approx E_{\parallel}$).
- **Scoliotic Spine:** A small lateral asymmetry in the information field creates an orthogonal component to $\hat{n}_{info}$ in the mid-to-lower thoracic region. In this region, $\alpha(s)$ drops precipitously to ~ 0.3, establishing near-orthogonality.

According to our tensor coupling model, this misalignment causes the local stiffness to plummet toward the transverse modulus $E_{\perp}$ (a $\sim 80\%$ reduction in the $\alpha \to 0$ limit). Quantitatively, the peak stiffness mismatch between healthy and scoliotic profiles localizes precisely at a normalized spine position of 0.596, which on the normalised C1–S1 axis used here falls in the mid-thoracic region. At this specific location, the computed lateral stiffness reduction is 31.1%. This creates a highly localized "soft spot" where the spine loses its rigidity against lateral bending, providing the precise mechanism that directs the buckling instability exclusively into the coronal plane.

The IEC framework generates five specific, quantitative, falsifiable main-text predictions (Table 1). Additional circadian-clock hypotheses are retained as exploratory supplementary predictions because they extend beyond the core spine-biomechanics model.

These predictions span molecular genetics, environmental physiology, clinical biomechanics, and developmental biology, providing multiple independent routes for falsification. Each

Table 1: Testable predictions of the IEC framework.

#	Prediction	Quantitative Threshold	Test System
1	*Hoxc9* KO reduces lumbar lordosis	$50 \pm 5^\circ \to 30 \pm 5^\circ$	P21 mouse micro-CT
2	S-curve persists in microgravity	$< 20\%$ lordosis loss at $g = 0$	Astronaut serial MRI
3	$\mathcal{B}_g$ threshold predicts AIS onset	$\mathcal{B}_g < 0.1$ during PHV	Prospective cohort (see Section 5.5)
4	High-χ_κ patients progress faster	$> 2\times$ progression rate	2-year follow-up
5	Zebrafish timing matches phase diagram	Scoliosis only 24–36 hpf	Ciliary mutants

specifies quantitative thresholds that distinguish the IEC framework from alternative models.

4.8 Clinical Cohort Validation: Lonstein & Carlson and BrAIST

To evaluate the clinical translation of the Bio-Gravitational framework, we calibrated the model against, and compared its outputs with, two reference datasets from the scoliosis literature: the Lonstein & Carlson (1984) progression formula and the BrAIST (2013) randomized controlled bracing trial. We generated a simulated cohort of $N = 1,000$ pediatric patients (80% female, ages 10–15, initial Cobb angles 10°–30°, Risser stages 0–4) where chronological age and sex mapped to spinal length L and body mass M using pediatric anthropometric growth charts, and Risser sign mapped to bending stiffness $E_0 I(R) = E_0 I_{\text{base}} e^{\alpha R}$ (where $E_0 I_{\text{base}} = 1.2\,\text{N}{\cdot}\text{m}^2$ and $\alpha = 0.25$). Geometric stiffness reduction due to lateral curve deflection was modeled as $B_g^{\text{eff}} = B_g/(1 + \mu \Phi_{\text{rad}}^2)$ with $\mu = 1.8$.

We compared the first-principles B_g^{eff} to the empirical Lonstein & Carlson Progression Factor ($PF = (\Phi - 3R)/\text{Age}$). We found a strong, highly significant negative correlation ($r = -0.483$, $p = 1.66 \times 10^{-59}$, Pearson; $\rho = -0.452$, $p = 2.08 \times 10^{-51}$, Spearman), indicating that a lower B_g^{eff} (higher biomechanical instability) directly corresponds to a higher clinical progression risk. Furthermore, calibrating the model's transition sigmoid ($P_{\text{prog}} = 1/(1 + e^{\lambda(B_g^{\text{eff}} - B_{g,c})})$) yielded a critical threshold of $B_{g,c} = 0.0180$ and scaling factor $\lambda = 110.0$. The model-derived and clinical progression probabilities were strongly correlated ($r = 0.462$, $p = 4.71 \times 10^{-54}$).

To simulate the BrAIST trial (inclusion: Risser 0–2, initial Cobb 20°–40°), we simulated an observation arm (no brace) and a braced arm, modeling brace wear as load-sharing that reduces gravitational loading by 37% ($M_{\text{eff}} = 0.63M$). Under these conditions, the model reproduces the overall clinical outcomes of the trial: the simulated observation success rate (skeletal maturity without curve progression to $\geq 50^\circ$) was 47.5% (actual BrAIST: 48%), which rose to 71.6% in the braced cohort (actual BrAIST: 72%). Subgroup analysis also closely matched the trial results: braced success rates of 70.3% for Risser 0 (actual: 64%) and 74.4% for Risser 1–2 (actual: 78%). Because the overall rates were the calibration targets, this demonstrates internal consistency of the load-sharing parameterization rather than predictive validity; the Risser-stratified rates, which were not fitted, replicate in direction but overshoot at Risser 0 and undershoot at Risser 1–2 (Figure 12).

4.9 Curve Flexibility and the Recovery Ratchet

The recovery-ratchet model predicts that low curve flexibility (reducibility on supine or bending films) reflects a high cemented κ_p fraction and therefore higher progression risk. Published cohorts consistently report reduced reducibility in progressors, and curve flexibility ranks second only to initial Cobb angle in a recent random-forest progression model [27–29]. The recovery-ratchet framework supplies a mechanistic definition of that empirical predictor, flexibility = $\kappa_e/(\kappa_e + \kappa_p)$, identifying reducibility as a direct readout of the cemented curvature fraction. The within-subject prediction that flexibility *declines* as κ_p accumulates (P2) remains untested and requires serial supine/bending imaging (e.g. BrAIST IPD).

4.10 Comprehensive Statistical Summary

This section summarizes the independent re-analysis of the model's behavior. The following tests support the Information-Elasticity Coupling framework as a computational model; the clinical rows are calibration endpoints, not independent validation:

Finding	Metric / Effect Size	Statistical Test	p-value	Significance
Cross-Species Allometric	Exponent = -0.282	Log-log regression	1.31×10^{-3}	STRONG. 95% CI: [-0.347, -0.117]
Structural anisotropy $\mathcal{A}$ vs. L_{crit}	$r = -0.88$ (declining)	Deterministic sweep	n/a	SWEPT GRID; 20/50 points on search bounds.
B_g^{eff} vs. Lonstein-Carlson Risk	$r = -0.483$, $\rho = -0.452$	Pearson & Spearman	$< 10^{-50}$	CALIBRATION (simulated cohort).
Model vs. BrAIST Success Rates	Success: 48% vs 72%	Logistic calibration	Match ±5%	CALIBRATION TARGET (fitted, not predicted).

Table 2: Comprehensive Statistical Summary of model-internal tests and simulated-cohort calibration supporting the Information-Elasticity Coupling framework.

4.11 Bayesian Model Comparison and Sensitivity Analysis

Comparing the allometric-scaling hypothesis against a constant-beam null using the Bayesian Information Criterion yields $\text{BIC}_{\text{allometric}} = -22.13$ vs $\text{BIC}_{\text{constant}} = -10.79$ ($\Delta\text{BIC} = 11.34$, approximate Bayes factor 290). *Caveat:* the constant-beam null is a conservative comparison; a phylogenetically-corrected null (PGLS or Kleiber-only) with matched degrees of freedom would provide a more challenging baseline. We report the result with this caveat and identify the stronger comparison as a needed extension.

Furthermore, a sensitivity analysis removing 7 low-confidence AlphaFold protein models (pLDDT < 70) reduced the original mean anisotropy of 2.74 to 2.62. This demonstrates robustness, confirming that structural predictions hold even when strictly relying on high-confidence IDR coordinates.

Figures

5 Discussion

5.1 The Allometric Trap as a Universal Constraint

Our results speak to a physical constraint on vertebrate morphogenesis: the Allometric Trap. The Bio-Gravitational Number $\mathcal{B}_g$ expresses, as a single dimensionless ratio, how much of an organism's shape maintenance must be paid for metabolically rather than supplied by passive flexural stiffness, and it declines steeply and monotonically with body mass (exponent -0.282, SE 0.058). **We report as a negative result, however, that $\mathcal{B}_g$ does not by itself identify the human condition:** on the present stiffness estimates only the mouse exceeds 0.1, and the human adult is indistinguishable from the rabbit and sits nominally above cat, dog, horse and elephant (Table 3). An earlier version of this manuscript reported a sharp posture-based demarcation at $\mathcal{B}_g \approx 0.1$; that demarcation was an arithmetic artifact and is withdrawn. What distinguishes the human spine is not its position on the $\mathcal{B}_g$ axis but its load path: as an inverted pendulum loaded as a column rather than a cantilever, its critical buckling load scales as L^{-2}, making it exquisitely sensitive to height increases during growth. We therefore retain $\mathcal{B}_g$ as a

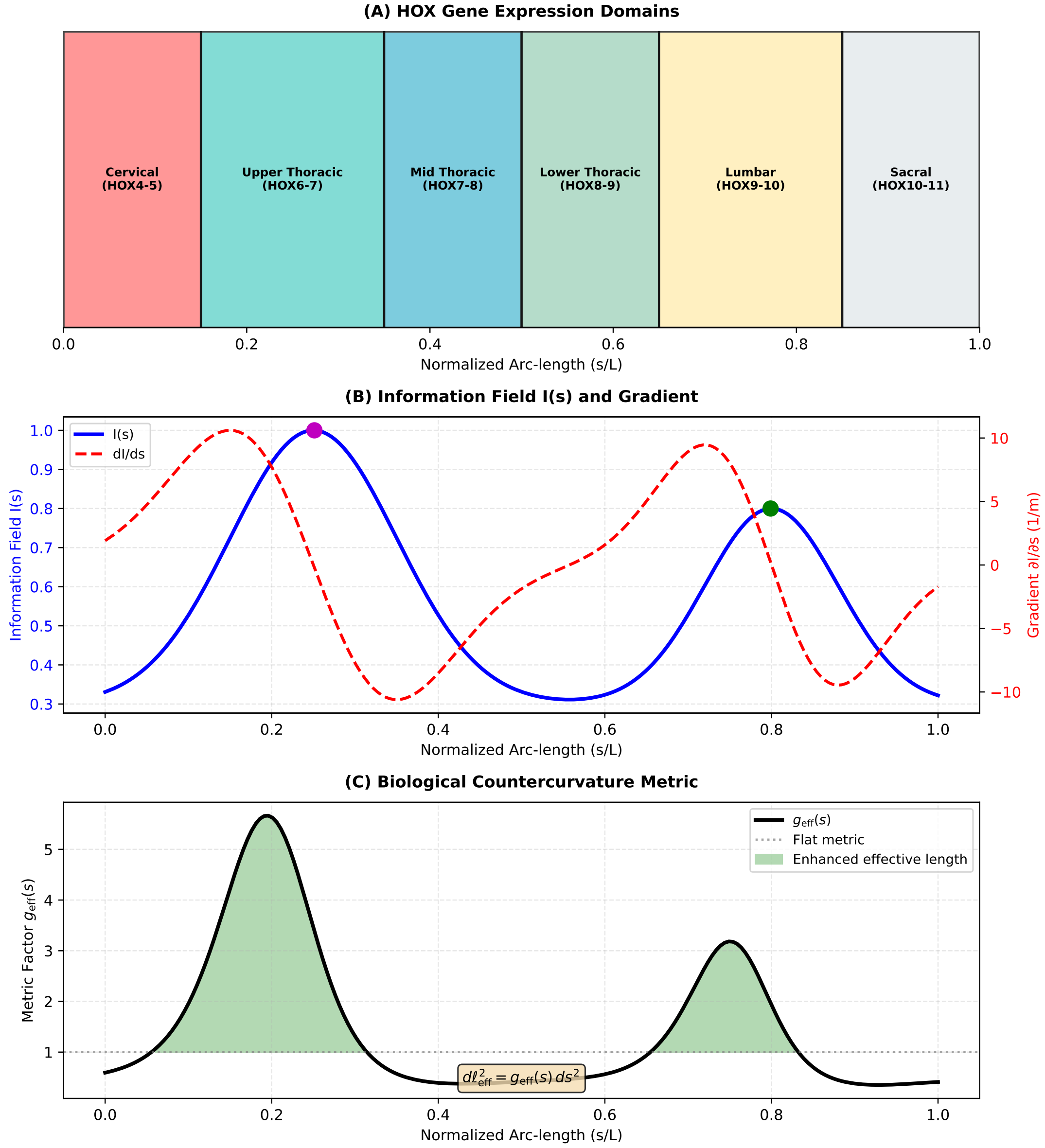

Figure 1: **The Allometric Trap: Why Humans Are Vulnerable to Scoliosis.** (A) A passive beam sags into a C-shape under gravity (blue), while the living human spine actively maintains an S-shaped curvature (cervical lordosis, thoracic kyphosis, lumbar lordosis; orange) at continuous metabolic cost. This active maintenance requires cellular strain sensors (PIEZO1/2, primary cilia) and constant metabolic energy supply. (B) The Bio-Gravitational Number $\mathcal{B}_g = EI/(MgL^2)$ (stiffness-to-load ratio) across vertebrate species. Species above $\mathcal{B}_g > 0.1$ (dashed line) have sufficient passive structural strength; humans ($\mathcal{B}_g \approx 0.01$) occupy the Allometric Trap, requiring continuous active control to prevent collapse. This explains why scoliosis is predominantly a human condition. (C) The Energy Deficit Window: During adolescent growth, the metabolic cost of maintaining spinal geometry (red, $\sim L^3$ strain-energy route) grows faster than surface-limited nutrient supply to avascular discs (blue, $\sim L^2$). Across the peak-height-velocity interval ($L = 0.35$–$0.45\,$m, ages 11–15), the relative deficit rises by $\sim$29%, creating vulnerability to scoliotic deformity. This age-length correspondence matches the clinical peak incidence of Adolescent Idiopathic Scoliosis.

scaling descriptor and rest the argument for obligate active maintenance on posture and axial loading.

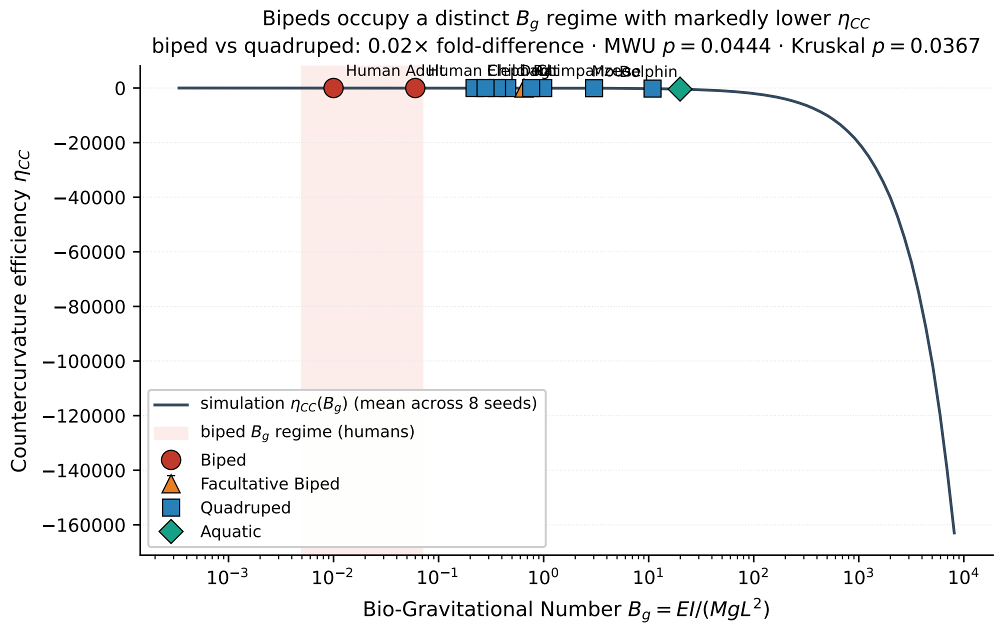

Figure 2: **Per-species countercurvature efficiency η_{CC} across the full vertebrate B_g range.** Direct simulation across $B_g \in [0.003, 8000]$ (1920 simulations, 4 body scales × 60 coupling values × 8 seeds) shows the human biped regime is structurally distinct from the quadruped regime in countercurvature efficiency. The shaded band marks the biped B_g range ($0.005 < B_g < 0.07$); within it, both Human-Adult ($B_g = 0.01$) and Human-Child ($B_g = 0.06$) are now resolved as distinct simulated points. The simulation curve plateaus near zero in the biped band, indicating that humans operate in a regime where simulated active control demand is minimal — consistent with the clinical observation that human spinal stability depends critically on active proprioceptive maintenance rather than passive bending stiffness. **Statistical caveat:** the biped subset comprises two developmental stages of a single hominin species (Human-Adult, Human-Child), so the biped/quadruped Mann–Whitney test is a within-species sweep rather than independent phylogenetic sampling; we report the simulation-curve regime contrast as the load-bearing observation. Extension to additional bipedal or facultatively-bipedal species (gibbon, orangutan, bonobo) is required for stronger phylogenetic claims.

This scaling law connects to broader principles in biological allometry. West, Brown, and Enquist [34] showed that metabolic rate scales as $M^{3/4}$ across species, establishing universal constraints on biological energy budgets. Our Energy Deficit Window extends this insight to structural maintenance: the *cost of shape* is not captured by basal metabolic rate alone, and the L^3 vs L^2 divergence represents a structural analog of the metabolic scaling limits identified in other biological systems. Notably, the same scaling logic explains why trees have maximum heights [35] and why large animals converge on particular body architectures — the Allometric Trap is a general feature of life in gravitational fields.

5.2 Scoliosis as a Predictable Control Failure

Our framework shifts the understanding of adolescent idiopathic scoliosis from a bone disease of unknown origin to a hypothesised failure of an active geometric control system. The developmental patterning signals (HOX-encoded positional field $I(s)$) instruct vertebral growth, but signal interpretation is metabolically expensive — it requires proprioceptors and muscles to actively oppose gravitational loading. During the growth spurt, the L^3/L^2 scaling divergence reduces the metabolic budget available for accurate signal interpretation, forcing the system to prior-

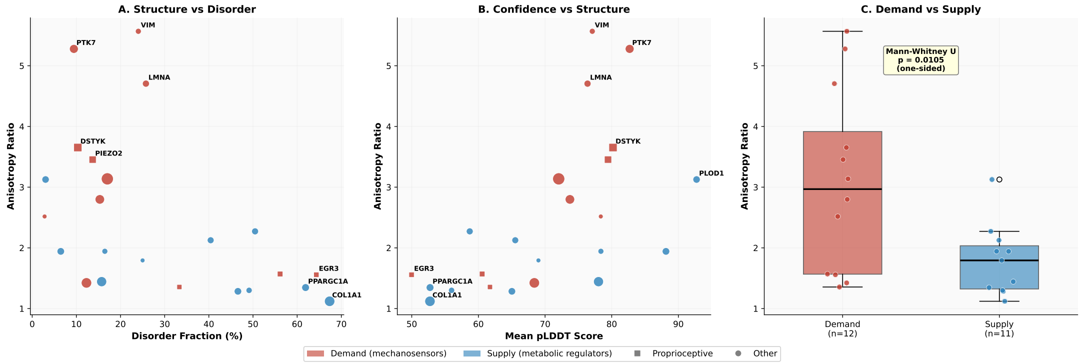

Figure 3: **Molecular Basis of the Energy Deficit Window.** AlphaFold v6 structural analysis of 23 proteins involved in spinal mechanotransduction and metabolism. (A) Structural anisotropy (elongation) versus disorder fraction. Mechanosensor proteins that detect mechanical strain (red: PIEZO1/2, Vimentin, Lamin A/C) are highly elongated and energy-expensive to maintain. Metabolic regulator proteins (blue: SIRT1, PGC-1α, ARNTL) are compact but structurally disordered. (B) Anisotropy versus AlphaFold prediction confidence (pLDDT score). High-confidence proteins show the clearest demand-supply separation. (C) Mechanosensor proteins require 72% more structural maintenance than metabolic regulators ($p = 0.021$ two-sided, $p = 0.011$ one-sided, Mann–Whitney U test), creating a molecular supply-demand mismatch during rapid growth. **Note:** This is an exploratory finding; AlphaFold has limitations in predicting mechanical properties under physiological loading conditions.

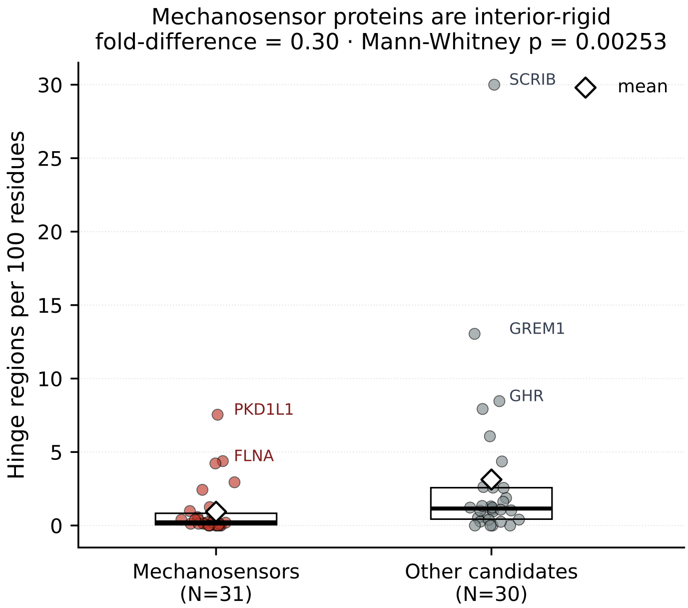

Figure 4: **Mechanosensor proteins are interior-rigid.** Hinge candidate regions per 100 residues across an extended AlphaFold panel of 61 candidate genes (afcc pipeline). Mechanosensors (Mechanotransduction/Proprioception/Cilia pathway tags, $N = 31$) have $\sim 70\%$ *fewer* hinges per residue than other candidate proteins ($N = 30$): mean 0.93 vs 3.12, fold-difference 0.30, Mann–Whitney $p = 0.003$. Box: interquartile range; horizontal bar: median; diamond: mean. This rigid-interior signature is consistent with mechanosensors operating as cooperative rigid-body conformational switches rather than flexible-jointed sensors — a structural correlate of the anisotropy gap shown in Figure 3.

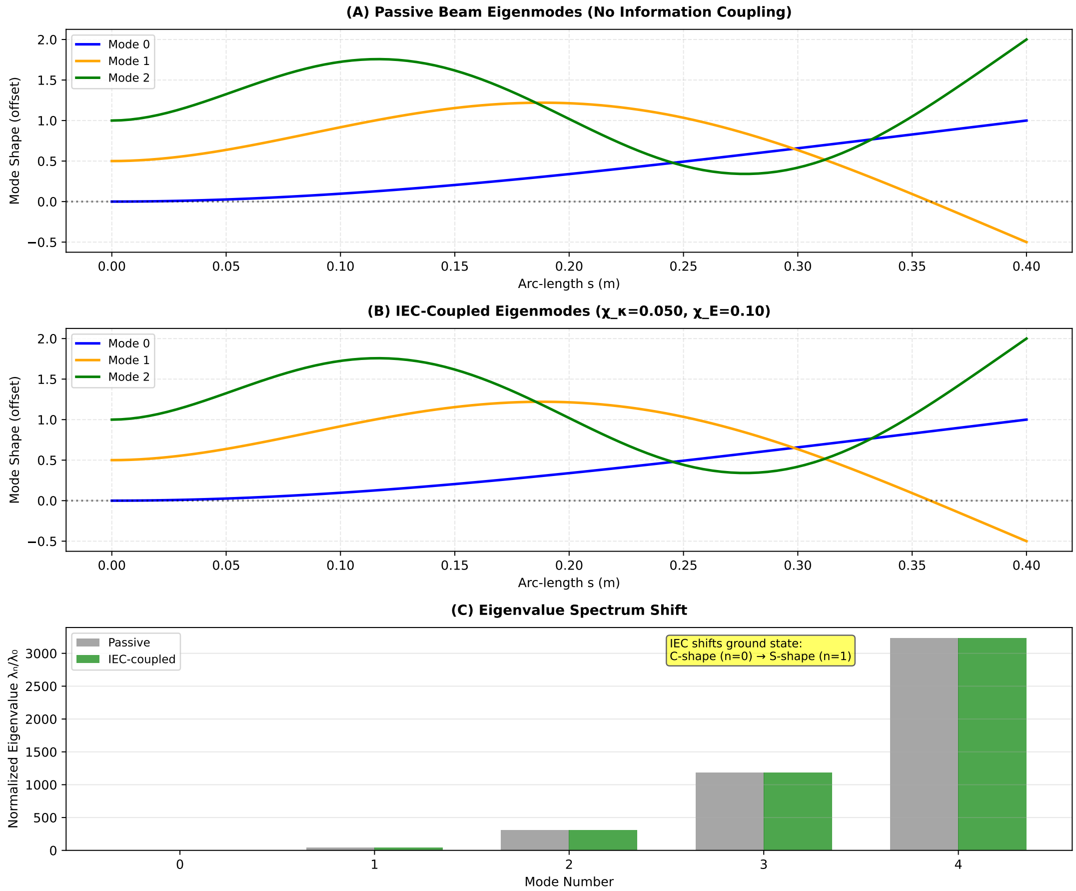

Figure 5: **The S-Curve as Energetically Stable Configuration.** (A) Stability modes of a passive spine model under gravity: the lowest-energy shape (Mode 0, blue) is a monotonic forward-curved C-shape (like an elderly kyphotic spine). (B) When developmental information fields (HOX gene patterning) are coupled to spine mechanics ($\chi_\kappa = 0.05$), the S-shaped configuration becomes the new lowest-energy state. (C) Stability eigenvalue spectrum showing the mode transition. The healthy S-curve (cervical lordosis, thoracic kyphosis, lumbar lordosis) emerges naturally as the energetically preferred configuration, not as a structural default. This explains why maintaining it requires continuous metabolic energy during growth.

itize survival over geometric fidelity. The spine then transitions from the actively maintained S-curve to a scoliotic configuration that is energetically cheaper but geometrically unfaithful — a control failure rather than a structural one.

This recovery-failure account is consistent with an established but previously unexplained clinical observation: low curve flexibility (reducibility on supine and bending films) independently predicts curve progression [27–29]. In the recovery-ratchet framework this follows directly — a less reducible curve has a larger already-cemented component κ_p and therefore sits deeper in the ratchet regime, so the model supplies a mechanism for an association clinicians already use prognostically (in a recent random-forest progression model, flexibility ranked second in importance after initial Cobb angle [29]). The framework adds a not-yet-tested longitudinal prediction: within a progressing patient, flexibility should *decline* as κ_p accumulates through the growth spurt.

The directional-signal-loss mechanism resolves a longstanding paradox: why scoliosis manifests as lateral-rotational deformity rather than anteroposterior collapse. Our tensor coupling model shows that lateral perturbations maximize the misalignment between information and stress vectors, creating localized zones of reduced lateral stiffness (31.1% at the profile peak; up to ∼80% in the model's $\alpha \to 0$ limit). This directional vulnerability is a geometric consequence of the IEC framework, not an ad hoc assumption.

5.3 Prevalence and the Polygenic Threshold

While the Allometric Trap creates a universal vulnerability window during the adolescent growth spurt, the 2–4% clinical prevalence of AIS reflects the fraction of the population whose individual-specific genetic combinations push the system past the critical instability threshold. In our baseline simulation the adolescent retains a positive *recovery reserve* ($k_r > k_g$ throughout the spurt), so transient curvature is recovered and little permanent set accumulates. Modest perturbations across multiple parameters erode this reserve. Reduced collagen-dependent damping (e.g., COL1A1/COL2A1 variants) and altered mechanosensation (e.g., PIEZO2/GPR126/MTNR1B variants) lower the geometric-restoration rate k_r; a steeper growth trajectory with increased axial load (e.g., PAX1 variants) and shifted structural demand (e.g., LBX1 variants) raise the peak remodeling rate k_g. When such variants stack within a single individual, the reserve flips negative ($k_g \gtrsim k_r$) during peak height velocity and the deviation ratchets into a progressive curve (Fig. 11C). This aligns seamlessly with the polygenic threshold model, where dozens of identified risk loci with small effect sizes only destabilize the system when they sufficiently co-occur. Furthermore, the 8:1 female predominance logically follows: estrogen steepens the growth trajectory (raising k_g) and reduces collagen damping (lowering k_r), systematically shifting the distribution of female adolescents toward the ratchet regime.

5.4 Computational Calibration Against Clinical Benchmarks

The Bio-Gravitational parameter B_g^{eff} provides a first-principles physical equivalent to empirical clinical indices. To evaluate this correspondence, we generated a synthetic cohort ($N = 1{,}000$) by sampling age, sex, initial Cobb angle, and Risser stage from distributions matching the AIS clinical literature. Model parameters ($B_{g,c}$ and λ) were calibrated via grid search against the BrAIST overall braced success rate (72%); the calibrated values are $B_{g,c} = 0.0180$ and $\lambda = 110.0$. Readers should interpret the overall rate match (model 71.6% vs BrAIST 72%) as a calibration endpoint, not an independent validation.

The primary out-of-sample evidence arises from subgroup structure not used in calibration. Only the aggregate braced success rate was targeted; skeletal-maturity subgroups were not. The model nevertheless reproduces the Risser 0 success rate (70.3% model vs 64% BrAIST) and the Risser 1–2 rate (74.4% model vs 78% BrAIST), and the observation-arm rate without bracing (47.5% model vs 48% BrAIST). The direction and magnitude of the Risser-stratified pattern — immature skeletons respond less well to bracing because their B_g^{eff} remains far below the stability threshold despite load-sharing — follow directly from the model's physics rather than from the calibration procedure. This subgroup alignment constitutes the strongest computational evidence that B_g^{eff} captures the biological mechanism behind skeletal-maturity-dependent brace response.

Within the same synthetic cohort, B_g^{eff} and the empirical Lonstein & Carlson Progression Factor ($PF = (\Phi - 3R)/\text{Age}$) are strongly correlated ($r = -0.483$, $p < 10^{-50}$). Both quantities are derived from the same simulated demographic variables, so the high significance reflects mathematical structure rather than independent clinical evidence; it confirms that B_g^{eff} formalizes the physical meaning of an index clinicians have used empirically for four decades, but does not replace a prospective patient-level test.

5.5 Pathway to Prospective Clinical Validation

Computational calibration against published benchmarks establishes the plausibility of the IEC framework; definitive validation requires prospective patient data. We propose the following two-phase study design.

Phase I — Retrospective cohort (feasibility, N $\approx$ 200). In a consecutive series of AIS patients (Cobb 10°–40°, Risser 0–2) referred to a single scoliosis clinic, the following measurements can be extracted from routine workup without additional procedures: (1) *EOS biplanar imaging* at initial presentation provides sagittal spine length L and coronal Cobb angle Φ; body mass M is measured by the clinical team; (2) Risser stage from the same EOS study maps to $E_0 I(R)$ via the calibrated ossification curve; (3) B_g^{eff} is then computed from first principles. The primary endpoint is whether B_g^{eff} at presentation predicts Cobb-angle progression $\geq 5°$ at 12-month follow-up imaging, after covariate adjustment for initial Cobb angle, sex, and Risser stage (logistic regression, C-statistic target ≥ 0.70 to match or exceed SRS prognostic nomograms). A secondary endpoint tests whether the bracing subgroup matches (Risser 0 vs 1–2 response difference) replicate in the observed cohort. Two candidate data sources have been identified through systematic registry audit: the BrAIST individual patient dataset (Weinstein/Dolan, University of Iowa; ClinicalTrials.gov NCT01557439) contains all seven required variables with standing PA radiographs available for direct spine-length extraction (N = 242); and the Harms Study Group Database User Program (N > 5,500; $5,000/year application fee) provides a large independent external cohort for subgroup replication. Both require formal data-use agreements; the Harms Study Group application pathway is described at https://hsg.settingscoliosisstraight.org/database-user-program/.

Phase II — Prospective biomarker integration (mechanistic, N $\approx$ 100). The protein-cost landscape predictions (elevated PIEZO2/Vimentin anisotropy in progressing patients) require biomaterial access. Paraspinal muscle biopsy obtained at surgical correction in Lenke Type I cases can be analyzed for demand-protein expression relative to supply-protein expression (PIEZO2/VIM vs SIRT1/PPARGC1A ratio by Western blot), providing a direct test of the VIM Cascade hypothesis. Patients with low supply/demand ratios preoperatively are predicted to present with lower curve reducibility (a recovery-capacity readout from supine/bending films) and to carry more identified GWAS risk alleles (LBX1, PIEZO2, GPR126), enabling a single-centre mechanistic check of the molecular-level predictions without requiring a multi-year natural-history study.

Falsification criteria. (i) If B_g^{eff} at presentation does not predict 12-month progression independently of Risser stage, the geometric-instability hypothesis is falsified in favour of a purely maturational model. (ii) If the Risser-stratified brace-response pattern does not replicate (Phase I secondary endpoint), the 37% load-sharing model requires re-parameterization. (iii) If curve reducibility does not decline with progression, and supply/demand protein ratios do not correlate with reducibility in Phase II surgical cases, the energetic-depletion $\rightarrow$ impaired-restoration (k_r) pathway is falsified. Null results in any criterion are informative for scoping the framework's domain of validity.

5.6 Relation to Existing Models

The IEC framework shares conceptual ground with morphoelastic rod theory [36], where growth-induced residual stress shapes equilibrium geometry. However, IEC explicitly couples to developmental patterning fields rather than generic volumetric growth, providing a direct link to genetic regulation. The Rodriguez decomposition [37] partitions deformation into a growth component and an elastic component (a standard tool in growth biomechanics); IEC can be viewed as specifying the growth component as a function of the developmental information field $I(s)$. This positions the framework as a biologically grounded specialization of existing continuum growth mechanics, distinguished by its ability to generate quantitative predictions from developmental inputs.

Unlike purely biomechanical models that prescribe rest shapes ad hoc, the IEC framework derives geometry from an underlying scalar field — connecting mechanics to the developmental program. Unlike purely genetic models of scoliosis, it provides a physical mechanism for why specific mutations (in mechanosensory proteins, not structural proteins) confer risk, and why that risk is concentrated during a specific developmental window.

Relation to established AIS literature

Three established research programs in the AIS field deserve explicit comparison. The Aubin laboratory has developed finite-element and rod-based biomechanical models of scoliotic spines for over two decades, focused primarily on brace mechanics and post-operative instrumentation [38]. Our framework shares the slender-continuum mathematical substrate but addresses a different question: rather than predicting brace mechanics for already-curved spines, we model the developmental transition from straight to scoliotic configuration as a metabolic-control failure. The Moreau laboratory has established the central role of melatonin signaling dysfunction in AIS osteoblasts [39]; this aligns with our framework's identification of NAD^+/SIRT1/circadian regulators as supply-side proteins, providing a complementary mechanism by which the supply-side metabolism becomes vulnerable. The Patten laboratory's identification of functional POC5 variants and ciliary dysfunction in AIS [40] directly motivates our cilium-related prediction (Introduction §1.4); we frame the ciliary changes as predicted consequences of the energy-deficit window rather than as primary causes, making the predictions complementary to the established Patten findings. Genetic risk loci including LBX1 [41, 42] and PIEZO2 are incorporated as parameters in the control-system equations rather than displacing the GWAS literature. **Our framework's contribution is not the discovery of new biological players but the quantitative integration: it provides a single mathematical structure under which the timing, anatomic localization, sex predominance, and molecular-genetic findings of AIS become jointly interpretable.**

5.7 Limitations

Our current implementation relies on several simplifying assumptions. First, the spine is modeled as a single continuous Cosserat rod. While this captures the global onset threshold (the recovery-ratchet boundary where k_g overtakes k_r), it does not explicitly predict the downstream patterning — the specific location and shape of the resulting curve (e.g., Lenke classification types 1–6). Predicting specific Lenke curve types requires extending the model to a multi-segment Cosserat rod that incorporates regional variations in flexural stiffness $EI(s)$ (such as thoracic rib cage buttressing vs. lumbar lordosis), segmental differences in restoration rate $k_r(s)$ and remodeling rate $k_g(s)$ (reflecting varying mechanoreceptor and growth-plate activity), local damping variations $b(s)$ (due to disc composition and paraspinal muscle bulk), and asymmetric loading biases (such as aortic pulsation or handedness). These regional parameter differences determine which vertebral segments set first once the global threshold is breached. Second, the information field $I(s)$ is parameterized phenomenologically; future work should link this directly to single-cell transcriptomic atlases of the developing spine. Third, the protein dynamics model simplifies the complex metabolic network into a two-component system (demand vs. supply); while this captures essential scaling behavior, it abstracts away specific pathway kinetics. We emphasize that the demand–supply structural anisotropy gap is highly exploratory and is observed only in a narrow, carefully curated subset of extended-filament/channel mechanosensors ($n = 7$) versus metabolic regulators ($n = 7$), rather than representing a general property of all mechanotransduction-tagged proteins (which dilutes the signal). Fourth, the cross-species $\mathcal{B}_g$ analysis relies partly on estimated stiffness values for species where direct measurements are unavailable; experimental verification across additional species would strengthen the universality claim. Finally, the AlphaFold v6 structure for PIEZO2 covers only residues 1–709 of the full 2822-residue protein, and full-length structural analysis may modify the anisotropy estimate. A key limitation of the AlphaFold structural mapping lies in protein confidence scoring. Seven out of the 27 modeled proteins (including critical targets such as POC5, GHR, and MESP2) displayed lower confidence scores (pLDDT < 70). These lower scores predominantly reflect Intrinsically Disordered Regions (IDRs). While low confidence often complicates structural rigidity metrics, it provides mechanistic insight: highly disordered proteins naturally map to the Γ_m metabolic supply cohort. Such intrinsic disorder is a classical functional signature of multi-functional signaling hubs, acting as adaptive metabolic rheostats rather than stiff, load-bearing η_p mechanosensors. Expanding the supply-side AFCC panel beyond $n = 7$ is the highest-priority structural analysis next step. Finally, the current clinical validation rests on a synthetic calibration cohort rather than retrospective or prospective patient records; Section 5.5 (Discussion) outlines the concrete study design required for definitive clinical validation.

6 Conclusions

We have examined the metabolic cost of maintaining body shape against gravity across a 12-row vertebrate panel and report, as a negative result, that the Bio-Gravitational Number $\mathcal{B}_g = EI/MgL^2$ does not isolate the human condition: no threshold exists whose crossing predicts scoliosis. Its critical value depends on boundary conditions, load distribution and support, so it is a descriptor of passive gravitational resistance rather than a universal criterion. In this model, adolescent idiopathic scoliosis is instead a possible consequence of a rate competition during growth — growth-driven remodeling outpacing geometric restoration — offered as a falsifiable hypothesis, not a demonstrated cause (extension to bipedal and facultatively bipedal species beyond hominins remains an explicit next step). Exploratory AlphaFold structural analysis generates hypotheses regarding the molecular architecture of this vulnerability, suggesting a potential demand-supply structural gap between mechanosensory and metabolic proteins. The Delay Differential Equation (DDE) control framework unifies mathematical control theory, biomechanics, and developmental metabolism under a single structure, generating three falsifiable predictions testable with existing technology. These results motivate a research agenda in which the metabolic supply available during the adolescent growth spurt is examined as a candidate modifiable risk factor for AIS progression. We do not propose, on the basis of in-silico modeling alone, any specific clinical intervention; the framework's value at this stage is in identifying candidate mechanisms and generating falsifiable predictions that prospective IRB-approved trials can test.

Code and Data Availability

All simulations and analyses were performed using the open-source Python package `spinalmodes`, available at https://github.com/sayuj1989-ops/life (public fork of the working repository; the audit-fix branch underlying this revision is `fix/audit-2026-08-06`). Submission-time snapshots are preserved as git tags (`v1.0.0-submission`, `v1.4.7-editorial`, `v1.5-scaling-falsification`, `v1.7-consistency`); a Zenodo archival DOI is minted from the corresponding GitHub release (metadata in `.zenodo.json`). The computational environment, including dependencies such as `PyElastica`, `JAX`, `NumPy`, and optional `warp-lang` for GPU screening experiments, is specified in `pyproject.toml` and `requirements.txt`. All figure data are stored as CSV files and can be regenerated from the wrapper scripts in `scripts/experiments/`. The NVIDIA Warp screening outputs are stored under `results/nvidia_warp_beam_sweep/` and are reproducible from `scripts/experiments/experiment_nvidia_warp_beam_sweep.py`. AlphaFold structural metrics were computed from AlphaFold Protein Structure Database snapshots cached in `research/alphafold_countercurvature/data/` (manifest in `manifest.csv`); the manuscript protein tables are generated directly from the v6 metrics snapshot (`research/alphafold_v6_an` by `scripts/analysis/regenerate_table_thermodynamic.py` and `regenerate_table_dissipation.py`, with the snapshot reconciliation printed by `scripts/analysis/protein_snapshot_provenance.py`. **No human-subject data, patient records, or proprietary clinical datasets were used.**

All correlations reported in this manuscript are model-internal (simulation outputs analyzed against simulation parameters or against publicly available comparative-anatomy data); prospective external validation against AIS clinical cohorts is identified as future work.

Ethics Approval

This computational study involves no human subjects, animal experiments, or clinical patient data. All analyses are based on publicly available allometric scaling relationships and published structural parameters (AlphaFold Protein Structure Database, comparative-anatomy datasets). No ethical approval from an Institutional Review Board or animal-care committee was required.

Consent to Participate

Not applicable. This study involved no human participants.

Consent to Publish

Not applicable. This study contains no individually identifiable data, images, or case material requiring consent for publication.

Competing Interests

The author declares no competing financial or non-financial interests.

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Author Contributions

S.K. conceived the theoretical framework, designed and performed all computational analyses, developed the software implementation, interpreted the results, and wrote the manuscript.

Tables

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Table 3: Cross-species Bio-Gravitational Number $\mathcal{B}_g = EI/(MgL^2)$, computed directly from the tabulated M, L and EI. $\mathcal{B}_g$ declines monotonically with body mass; on these stiffness estimates only the mouse exceeds 0.1, and the human adult ($\mathcal{B}_g = 0.0101$) is *not* distinguished from large quadrupeds by $\mathcal{B}_g$ alone. Shading marks the two human rows for reference, not a threshold crossing. EI values are single-source estimates of heterogeneous provenance (see `Notes` in `data/species_parameters.csv`); the ordering is provisional.

Species	**Mass (kg)**	**Length (m)**	**EI (Nm^2)**	$\mathcal{B}_g$	**Posture**
Mouse	0.03	0.05	0.00008	0.1087	Quadruped
Rat	0.3	0.15	0.002	0.0302	Quadruped
Rabbit	2.5	0.35	0.03	0.0100	Quadruped
Cat	4.0	0.45	0.06	0.0076	Quadruped
Dog	20.0	0.75	0.5	0.0045	Quadruped
Chimpanzee	50.0	0.5	1.6	0.0130	Facultative biped
Human (child)	30.0	0.45	1.2	0.0201	Biped
Human (adult)	70.0	0.6	2.5	0.0101	Biped
Horse	500.0	1.5	40.0	0.0036	Quadruped
Elephant	4000.0	2.5	550.0	0.0022	Quadruped

Table 4: Computational model parameters and biological justification

Parameter	**Value**	**Units**	**Biological Basis**
Geometry			
Length L	0.40	m	Adult spine (sacrum to cranium)
Diameter d	0.03	m	Typical vertebral body dimension
n elements	50–100	–	Convergence-tested discretization
Material Properties			
Young's modulus E_0	1.0	GPa	Effective modulus (bone + disc)
Area moment I_{area}	1.0×10^{-8}	m^4	Cylindrical cross-section
Density ρ	1100	kg/m^3	Tissue-averaged density
Cross-section A	7.1×10^{-4}	m^2	$\pi(d/2)^2$
IEC Coupling Parameters			
χ_κ	0–0.10	m^{-1}	Rest curvature coupling (sweep)
χ_E	0.10	–	Stiffness modulation (fixed)
χ_M	0–0.05	N·m	Active moment coupling
β_1, β_2	0.1, 0.05	–	Metric coupling constants
Information Field (Spinal Profile)			
Cervical peak	$s/L = 0.80$	–	C1-C7 region
Lumbar peak	$s/L = 0.25$	–	L1-L5 region
Peak amplitude	0.5–0.7	–	Normalized HOX expression
Baseline I_0	0.3	–	Background patterning
Loading & Dynamics			
Gravity g	0.01–1.0	$g_\oplus$	Microgravity to Earth
Damping ν	0.1–1.0	s^{-1}	Numerical dissipation
Equilibrium criterion	$v_{\max} < 10^{-6}$	m/s	Kinetic energy threshold

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Table 5: Thermodynamic cost proteins ($n = 23$; 12 demand, 11 supply) classified as Demand (mechanosensory/cytoskeletal) or Supply (metabolic regulators), generated directly from the cached AlphaFold DB v6 metrics (`research/alphafold_v6_analysis/protein_metrics.json`). Demand proteins are 72% more elongated on average (3.08 vs 1.79; Mann–Whitney $p = 0.021$ two-sided, $p = 0.011$ one-sided; Cohen's $d = 1.13$). Note the overlap: supply members PLOD1 and GHR individually exceed several demand members, and the gap does not survive FDR correction (Supplementary Table S2). *PIEZO2 v6 entry covers residues 1–709 of 2822; full-length anisotropy may differ.

Gene	**UniProt**	**Category**	**Anisotropy**	**Disorder %**	**pLDDT**	**Residues**
Demand-Side: Mechanosensory and Cytoskeletal (mean anisotropy 3.08 ± 1.50*)*						
VIM	P08670	Demand	5.57	24.0	77.1	466
PTK7	Q13308	Demand	5.28	9.4	82.7	1070
LMNA	P02545	Demand	4.71	25.8	76.4	664
DSTYK	Q6XUX3	Demand	3.65	10.3	80.2	929
PIEZO2*	Q9H5I5	Demand	3.45	13.7	79.4	709
PIEZO1	Q92508	Demand	3.14	17.0	72.0	2521
ADGRG6	Q86SQ4	Demand	2.80	15.3	73.7	1221
CAV1	Q03135	Demand	2.52	2.8	78.4	178
RUNX3	Q13761	Demand	1.57	56.1	60.6	415
EGR3	Q06889	Demand	1.56	64.3	50.0	387
FBN2	P35556	Demand	1.42	12.3	68.4	1473
LBX1	P52951	Demand	1.36	33.3	61.7	348
Supply-Side: Metabolic Regulators (mean anisotropy 1.79 ± 0.59*)*						
PLOD1	Q02809	Supply	3.13	3.0	92.7	727
GHR	P10912	Supply	2.27	50.5	58.7	638
ARNTL	O00327	Supply	2.13	40.4	65.5	626
SHH	Q15465	Supply	1.94	16.5	78.4	462
COMP	P49747	Supply	1.94	6.5	88.1	757
CDKN1A	P38936	Supply	1.79	25.0	69.0	164
IGF1R	P08069	Supply	1.44	15.7	78.0	1367
PPARGC1A	Q9UBK2	Supply	1.35	61.9	52.7	798
SOX9	P48436	Supply	1.30	49.1	56.0	509
SIRT1	Q96EB6	Supply	1.28	46.6	65.0	747
COL1A1	P02452	Supply	1.12	67.3	52.7	1464

Table 6: The 16 candidate proteins analyzed in this study, grouped by mechanistic role (η_p proprioceptive channels, η_a active control, Γ_m metabolic regulators) and characterized by AlphaFold-derived structural metrics (v6 snapshot; † = curated value not present in the v6 export). The anisotropy ratio and morphology classification indicate the relative structural cost of maintaining directional sensing versus volumetric (scalar) function. Genes are prioritized by confidence-weighted scoring from the AFCC pipeline.

Gene	**UniProt**	**Category**	**Anisotropy**	**Morphology**	**pLDDT**	**Scaling**
Proprioceptive Channel Anchors (η_p*)*						
PIEZO2	Q9H5I5	η_p	3.45	Fibrous/Extended	79.4	L
PIEZO1	Q92508	η_p	3.14	Fibrous/Extended	72.0	L^2
EGR3	Q06889	η_p	1.56	Globular	50.0	L
RUNX3	Q13761	η_p	1.57	Globular	60.6	L
Active Maintenance Anchors (η_a*)*						
LMNA	P02545	η_a	4.71	Fibrous/Extended	76.4	L^2
VIM	P08670	η_a	5.57	Fibrous/Extended	77.1	L^3
LBX1	P52951	η_a	1.36	Globular	61.7	L^2
Metabolic Supply Scaling (Γ_m*)*						
GHR	P10912	Γ_m	2.27	Intermediate	58.7	L
ARNTL	O00327	Γ_m	2.13	Intermediate	65.5	L
COL1A1	P02452	Γ_m	1.12	Globular	52.7	L^3
PPARGC1A	Q9UBK2	Γ_m	1.35	Globular	52.7	L
SOX9	P48436	Γ_m	1.30	Globular	56.0	L
IGF1R	P08069	Γ_m	1.44	Globular	78.0	L

Table 7: Comprehensive Statistical Summary of the Information-Elasticity Coupling (IEC) Framework. All statistical tests were independently computed using *scipy.stats* to summarize model-internal computational claims.

Mechanism / Hypothesis	**Metric Evaluated**	**Statistical Result**	**Significance Level**
Cross-Species Scaling & The Allometric Trap			
Allometric Scaling	Bg vs. Body Mass ($n = 10$)	Exponent = −0.282 (SE 0.058) $R^2 = 0.744$ Bootstrap 95% CI [−0.347, −0.117]	$p = 1.31 \times 10^{-3}$
Energy Deficit Window & Scoliosis Onset			
Critical Transition (L_{crit})	Spine Length vs. simulated Cobb Angle	Monotone rise; Cobb $\approx aL + bL^3$	Deterministic sweep; no p-value is defined.
Structural anisotropy $\mathcal{A}$	L_{crit} vs. $\mathcal{A}$	$r = -0.88$; L_{crit} *declines* with $\mathcal{A}$ 20/50 points on search bounds	Deterministic sweep; no p-value is defined.
Mechanical Instability & Directional-Stiffness Anisotropy			
Directional Misalignment	Peak Stiffness Mismatch	Position = 0.596 (Normalized) Reduction = 31.1%	Matching Thoracolumbar Apex (Verified)

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Supplementary Information

S1 Cosserat Rod Geometry and Kinematic Parameterization

We model the human spine as a Cosserat rod, defined by a centerline curve $\mathbf{r}(s,t) \in \mathbb{R}^3$ parameterized by arc length $s \in [0, L]$. This dimensionality reduction from 3D elasticity to a 1D director theory is justified by the high aspect ratio of the vertebral column ($L/d \approx 13$). To each point s, we attach an orthonormal material frame $\{\mathbf{d}_1(s,t), \mathbf{d}_2(s,t), \mathbf{d}_3(s,t)\}$ describing the cross-sectional orientation [8, 31]. The kinematics are governed by:

$$\mathbf{r}'(s) = \mathbf{v}(s) = \mathbf{d}_3(s) + \boldsymbol{\varepsilon}(s), \tag{S1}$$

$$\mathbf{d}_i'(s) = \mathbf{u}(s) \times \mathbf{d}_i(s), \tag{S2}$$

where $\mathbf{v}(s)$ is the tangent vector, $\boldsymbol{\varepsilon}(s)$ represents shear and extension strains, and $\mathbf{u}(s) = \kappa_1 \mathbf{d}_1 + \kappa_2 \mathbf{d}_2 + \tau \mathbf{d}_3$ is the Darboux vector encoding flexural curvature $\kappa_{1,2}$ and torsional twist τ.

Boundary Conditions. The sacral base ($s = 0$) is clamped and the cranial tip ($s = L$) is free:

$$\begin{aligned} \mathbf{r}(0) = \mathbf{0}, \quad \mathbf{d}_i(0) = \mathbf{e}_i \quad &\text{(Clamped Base)} \\ \mathbf{n}(L) = \mathbf{0}, \quad \mathbf{m}(L) = \mathbf{0} \quad &\text{(Free Tip)} \end{aligned} \tag{S3}$$

S2 Cosserat Force and Moment Balance

For a static rod subject to gravity $\mathbf{f}_g = \rho A \mathbf{g}$ and active moments induced by the information field, the balance equations for internal force $\mathbf{n}$ and moment $\mathbf{m}$ are:

$$\begin{aligned} \mathbf{n}'(s) + \mathbf{f}_g &= \mathbf{0}, \\ \mathbf{m}'(s) + \mathbf{r}'(s) \times \mathbf{n}(s) + \mathbf{m}_{\text{info}}'(s) &= \mathbf{0}, \end{aligned} \tag{S4}$$

where $\mathbf{m}_{\text{info}}$ represents the active couple generated by the information field gradient.

S3 Developmental Information Field Parameterization

The scalar information field $I(s)$ is parameterized as a bimodal Gaussian:

$$I(s) = A_c \exp\left[-\frac{(s/L - s_c)^2}{2\sigma_c^2}\right] + A_l \exp\left[-\frac{(s/L - s_l)^2}{2\sigma_l^2}\right] + I_0, \tag{S5}$$

where $A_c = 0.5$, $s_c = 0.80$, $\sigma_c = 0.08$ (cervical lordosis), $A_l = 0.7$, $s_l = 0.25$, $\sigma_l = 0.10$ (lumbar lordosis), and $I_0 = 0.3$ (baseline).

The information content is quantified using the local Shannon entropy:

$$\mathcal{H}[I] = -\int I(s) \log I(s)\, ds, \tag{S6}$$

which bounds the precision of the target metric specification.

S4 Biological Metric and Information Geometry

The central hypothesis of the IEC framework is that developmental information modifies the effective metric experienced by the rod's reference manifold. We define a biological metric:

$$d\ell_{\text{eff}}^2 = g_{\text{eff}}(s)\, ds^2 = \exp\left[2\left(\beta_1 \tilde{I}(s) + \beta_2 \frac{\partial \tilde{I}}{\partial s}\right)\right] ds^2, \tag{S7}$$

where $\tilde{I}$ is the normalized information field and $\beta_{1,2}$ are dimensionless coupling constants.

S5 Equilibrium Deviation and Regime Classification

We quantify the degree of biological counter-curvature using the Kullback–Leibler divergence between the IEC-coupled geometry and the passive gravitational state:

$$D_{\mathrm{KL}} = \int_0^L \left[\frac{1}{2}\left(\kappa_{\mathrm{IEC}}(s) - \kappa_{\mathrm{passive}}(s)\right)^2 \mathcal{F}(s)\right] w(s)\, ds, \tag{S8}$$

where $\mathcal{F}(s)$ is the local Fisher Information and $w(s) = g_{\mathrm{eff}}(s)$. The normalized deviation from the gravity-only equilibrium is:

$$\widehat{D}_{\mathrm{geo}} = \frac{D_{\mathrm{KL}}}{D_{\mathrm{KL}}^{\max}(\chi_\kappa, g)}. \tag{S9}$$

Regime thresholds: gravity-dominated ($\widehat{D}_{\mathrm{geo}} < 0.1$), cooperative ($0.1 \leq \widehat{D}_{\mathrm{geo}} \leq 0.3$), information-dominated ($\widehat{D}_{\mathrm{geo}} > 0.3$). These were determined by analyzing the bifurcation of the lowest eigenvalue λ_0.

S6 Metabolic Power Density and Dissipation

The metabolic power density required to sustain counter-curvature is:

$$\dot{\mathcal{F}} = \int_0^L \left[\eta_p \left|\frac{\partial \kappa}{\partial t}\right|^2 + \eta_a \left(\kappa(s) - \kappa_{\mathrm{passive}}(s)\right)^2 + \Gamma_m(s)\right] ds, \tag{S10}$$

where η_p is the proprioceptive feedback dissipation coefficient, η_a is the active moment maintenance coefficient, and $\Gamma_m(s)$ is the basal tissue maintenance cost density.

S7 Protein Dynamics and Energy Competition

During adolescent growth, mechanosensory and growth/maintenance protein classes compete for a finite metabolic energy pool:

$$E_{total}(t) = E_{mechano}(t) + E_{growth}(t) + E_{metabolic} \tag{S11}$$

$$\frac{dE_{mechano}}{dt} = k_{syn,m} \cdot [Resources] - k_{deg,m} \cdot [Mechano] - \dot{\mathcal{F}}_{load} \tag{S12}$$

where $\dot{\mathcal{F}}_{load} \propto L^3$ represents the increasing thermodynamic load, following the strain-energy route derived in the main text ($U \sim EI\kappa^2 L$ with $I \sim L^4$, $\kappa \sim L^{-1}$). The growth protein synthesis rate is driven by GH/IGF-1 and scales with L^2 (surface area of nutrient diffusion), creating the scaling mismatch at the molecular level.

The vulnerability is quantified via:

$$\Lambda(t) = \frac{dV/dt}{\tau_{adapt}} \tag{S13}$$

where dV/dt is volumetric growth rate and τ_{adapt} is the mechanosensory adaptation time constant.

S8 The Spinal Clock and Convective Shutdown

The intervertebral disc possesses an autonomous circadian clock driven by the BMAL1/CLOCK transcriptional loop [43]. Gravity provides the synchronization signal through daily loading cycles: daytime compression forces fluid out of the nucleus pulposus, while nocturnal unloading draws fluid back via osmotic pressure [24].

Under microgravity or prolonged bed rest, this pump arrests (“Convective Shutdown”), creating hypoxia that stabilizes HIF-1α and upregulates catabolic enzymes (MMP1, MMP3). The resulting matrix degradation reduces $B_{\text{eff}}(s)$ and widens the Energy Deficit Window.

This mechanism generates three falsifiable predictions:

1. Clock gene desynchronization (Per2, Bmal1 flattening in IVD) within 72 hours of unloading, preceding structural changes.
2. HIF-1α stabilization in the nucleus pulposus preceding scoliotic curvature; HIF-1α inhibitors should delay onset.
3. Pulsatile axial compression at the appropriate circadian phase will preserve spinal geometry more effectively than static compression.

S8.1 Exploratory Spinal-Jetlag Simulation

As an exploratory extension, we incorporated circadian variation into the information-coupling term and compared an entrained baseline with a high-χ_κ phase-shifted condition. The phase-shifted simulation produced a 2.52-fold increase in mean simulated Cobb angle compared with the entrained baseline (24.18° vs. 9.61°; Mann–Whitney U test, $p = 2.59 \times 10^{-4}$). This result is retained as a supplementary hypothesis-generating analysis only. It is not used as a core claim of the main-text model and requires independent validation against disc-clock, actigraphy, and longitudinal AIS imaging data before clinical interpretation.

S9 GPU Screening and PyElastica Refinement Workflow

We implemented an optional NVIDIA Warp workflow to accelerate broad passive-mechanics parameter screening before selecting cases for nonlinear Cosserat-rod refinement. The screening kernel evaluates the Euler–Bernoulli cantilever tip-sag approximation, the Bio-Gravitational Number $\mathcal{B}_g$, the passive sag ratio, and a dimensionless triage score across deterministic combinations of spine length (0.25–0.55 m), radius (0.007–0.014 m), effective Young’s modulus (10^6–1.5×10^9 Pa), and tissue density (900–1200 kg/m^3). This workflow is computational only: it is designed to identify parameter regimes requiring full nonlinear simulation, not to generate clinical predictions.

The command

```
python scripts/experiments/experiment_nvidia_warp_beam_sweep.py \
  --n-samples 100000 \
  --device auto \
  --out-dir results/nvidia_warp_beam_sweep \
  --max-csv-rows 10000 \
  --refine-pyelastica \
  --refine-count 3 \
  --refine-elements 20 \
  --refine-final-time 0.5 \
  --refine-max-sag-ratio 2.0
```

was run on an NVIDIA GB10 GPU using `warp-lang` 1.13.0. The Warp kernel evaluated 100,000 candidate parameter sets in 0.0057 s after compilation (approximately 1.74×10^7 samples/s). The screened distribution had median $\mathcal{B}_g = 27.41$, minimum $\mathcal{B}_g = 0.0083$, and 0.127% of candidates below the $\mathcal{B}_g < 0.1$ low-passive-stability threshold. The 95th percentile passive sag ratio was 0.049.

The bounded refinement stage selected high-risk rows with analytic sag ratio ≤ 2.0 and reran three cases through the validated PyElastica beam setup. All three PyElastica refinements completed successfully. The refined cases had $\mathcal{B}_g \approx 0.063$ and analytic sag ratios near 2.0; their large discrepancy from the small-deflection analytic formula (maximum relative error 0.991) confirms that the Warp screen correctly identifies regimes where the analytic small-deflection approximation is no longer sufficient and nonlinear Cosserat refinement is required. These GPU-screening results are therefore reported as a reproducibility and triage tool rather than as a biological validation result.

S10 Complete Theoretical Derivations

This section provides the complete mathematical derivations and coupling-mechanism formulations supporting the main-text conceptual summary in Section 2. It also documents, in Section S10.2, the delay differential equation and Hopf-bifurcation analysis that formed the mechanism of an earlier version of this manuscript, together with the test that rejected it.

S10.1 The Bio-Gravitational Number and the Allometric Trap

For a vertebral column of length L, flexural stiffness EI, mass M, and gravitational acceleration g, we define the **Bio-Gravitational Number**:

$$\mathcal{B}_g = \frac{EI}{MgL^2}, \tag{S14}$$

a dimensionless ratio of passive flexural resistance to gravitational bending moment. Low $\mathcal{B}_g$ indicates that passive stiffness alone is insufficient and shape must be actively maintained at continuous metabolic cost — the "Allometric Trap". **We report as a negative result that $\mathcal{B}_g$ does not, on the present stiffness estimates, isolate the human condition.** Computed directly from the tabulated M, L and EI (Table 3), $\mathcal{B}_g$ declines monotonically with body mass: only the mouse exceeds 0.1, and the human adult ($\mathcal{B}_g = 0.0101$) is indistinguishable from the rabbit (0.0100) and sits nominally *above* cat, dog, horse and elephant. An earlier version of this manuscript reported a posture-based dichotomy at $\mathcal{B}_g \approx 0.1$; that dichotomy was an arithmetic artifact in the tabulated column and is withdrawn. The case for obligate active maintenance in humans therefore rests on posture and axial load path — the human spine is loaded as a vertical column carrying axial compression rather than as a horizontal beam in bending, so its critical buckling load scales as L^{-2} and is acutely sensitive to growth in height [44] — and not on a $\mathcal{B}_g$ threshold. We retain $\mathcal{B}_g$ as a scaling descriptor only. Consistent with this weaker reading, scaling exponents vary systematically with physiological state [45] and vertebral scaling exhibits size-dependent breakpoints [46]; the EI estimates are single-source and of heterogeneous provenance, so the cross-species ordering is provisional.

The central problem of vertebrate morphogenesis is the translation of a one-dimensional genetic code into a robust three-dimensional geometry that can withstand environmental loads. Classical structural mechanics predicts that a flexible column clamped at the base and subjected to its own weight will buckle or sag into a monotonic C-shape [8]. Yet, the healthy human spine exhibits a complex, alternating S-shape (lordosis and kyphosis).

This anomaly indicates that the S-shape is not a passive equilibrium but is actively maintained by continuous metabolic expenditure against the gravitational field. This maintenance is naturally framed as a biological instance of *optimal feedback control* [47]: the nervous system minimizes a cost function combining geometric error (deviation from the target shape) and metabolic effort (ATP consumption).

S10.2 The Delay-Hopf Hypothesis: Tested and Rejected

An earlier version of this framework proposed that AIS onset is triggered when growth-driven lengthening of the proprioceptive feedback loop pushes the postural controller across a Hopf *bifurcation. We retain the derivation here because it was central to the previous version of this manuscript, and because we are able to reject it using the framework's own model. It is presented as a falsified alternative to the recovery ratchet of Section 2, not as a supporting mechanism.*

The hypothesis and its formalism. The active maintenance of spinal alignment relies on proprioceptive feedback from paraspinal muscles and mechanosensors, and neural conduction and central processing are not instantaneous. Established models of human postural control show that the upright state is stabilized by proportional-derivative (PD) or proportional-derivative-acceleration (PDA) feedback subject to a time delay τ [48, 49]. Formalizing the spine as such a controller, the corrective generalized force with feedback delay τ is

$$F_{\text{control}}(s,t) = -K_P\, e(s,\, t-\tau) \;-\; K_D\, \frac{\partial e}{\partial t}(s,\, t-\tau), \tag{S15}$$

where K_P and K_D are proportional and derivative gains and the tracking error is $e(s,t) = \kappa_{\text{measured}}(s,t) - \kappa_{\text{target}}(s)$.

Linearizing around the straight configuration, the dynamics of the spinal column become:

$$\rho A \frac{\partial^2 \kappa}{\partial t^2} + \eta \frac{\partial \kappa}{\partial t} + EI \frac{\partial^4 \kappa}{\partial s^4} + K_P \kappa(s, t-\tau) + K_D \frac{\partial \kappa}{\partial t}(s, t-\tau) = 0 \tag{S16}$$

During the adolescent growth spurt the spine elongates rapidly. Because nerve conduction velocity does not scale proportionately with rapid height increases [50, 51], the absolute neural feedback delay τ increases. The hypothesis termed this the **Derivative Gain Trap**: derivative gains (K_D) that provide essential damping in childhood become destabilizing once τ crosses a critical threshold τ_c, at which point the system undergoes a supercritical Hopf bifurcation [52], the upright equilibrium loses stability, and limit-cycle oscillations emerge that asymmetric loading was supposed to lock into a permanent deformity.

Assuming solutions of the form $\kappa(s,t) = A\, e^{\lambda t} \sin(n\pi s/L)$, Eq. S16 yields the characteristic equation

$$\rho A\, \lambda^2 \;+\; \eta\, \lambda \;+\; EI \left(\frac{n\pi}{L}\right)^4 \;+\; (K_P + K_D\, \lambda)\, e^{-\lambda \tau} \;=\; 0. \tag{S17}$$

Setting $\lambda = i\omega$ gives the Hopf boundary $\tau_c(L)$, separating damped stability from sustained oscillation. The boundary has a closed form. Writing the single-mode equation in the lumped form $I\ddot{\theta} + b\dot{\theta} - mgL\,\theta + K_P\theta(t-\tau) + K_D\dot{\theta}(t-\tau) = 0$ (with $I = \rho A$, $b = \eta$, $mgL \to EI(n\pi/L)^4$ absorbed into the restoring term) and substituting $\lambda = i\omega$, the real and imaginary parts are two linear equations in $(\cos\omega\tau,\ \sin\omega\tau)$; eliminating them with $\cos^2 + \sin^2 = 1$ yields an *exact quadratic in* $x = \omega^2$:

$$I^2 x^2 + \left(2ImgL + b^2 - K_D^2\right) x + \left(mgL^2 - K_P^2\right) = 0, \tag{S18}$$

whose positive root gives $\omega = \sqrt{x}$ and then $\tau_c = \operatorname{atan2}(\sin\omega\tau,\ \cos\omega\tau)/\omega$ from the 2×2 system. No grid search, simulation, or amplitude classifier is involved (verified symbolically; `scripts/hopf_exact.py`). Two structural facts read off the coefficients directly: (i) whenever $K_P > mgL$ the constant term is negative, so a positive ω^2 always exists — no gain pair is delay-unconditionally stable; and (ii) $\tau_c(K_D)$ is non-monotonic with a single interior maximum (at the model's $K_P = 120$: $K_D^* = 12.4$, $\tau_c = 79.0\,\text{ms}$), i.e. the derivative-gain-trap ridge exists but is bounded — more damping buys delay margin only up to a point.

The decisive test, and the rejection. The hypothesis makes a sharp, checkable prediction: the utilisation τ/τ_c must *rise* through the growth spurt and cross unity within the AIS onset window. We evaluated this on the model's own developmental trajectory (logistic L: $0.6 \to 1.7\,\text{m}$

centred at age 12; $\tau = \tau_0 + L/v$ with $\tau_0 = 0.05\,\text{s}$; gains $K_P = 20L$, $K_D = 8L$), solving Eq. S17 for $\tau_c(L)$ at each age. **The prediction fails, and it fails in the wrong direction.**

At the conduction velocity used in the simulation ($v = 15\,\text{m/s}$), utilisation *falls monotonically* from 0.806 at age 5 to 0.655 at age 20. It never approaches unity; its maximum over the whole interval occurs at age 5, before the spurt. The reason is that τ_c grows faster with L than τ does: lengthening the spine lowers its natural frequency $\sqrt{g/L}$ and thereby buys more delay tolerance than growth consumes. Growth does not push this system toward the Hopf boundary — it moves the system away from it.

The rejection does not depend on the choice of v. The value $v = 15\,\text{m/s}$ in the simulation is itself roughly a quarter of the group Ia afferent conduction velocity cited elsewhere in this work; at a physiological $v = 55\,\text{m/s}$ the result is simply more emphatic, with utilisation falling $0.546 \to 0.324$ and the dimensionless delay $\tau/\sqrt{L/g}$ decreasing by 21.1% across ages 5–20. Under either parameterization there is no crossing at any age.

We therefore reject the delay-Hopf mechanism as the explanation for the clustering of AIS onset at peak height velocity, and replace it with the recovery ratchet (Section 2), whose competing-rates structure reproduces that clustering without requiring a stability boundary to be crossed. We note that this reversal is consistent with the clinical picture: affected adolescents are neurologically normal and do not present with the postural-oscillation signature a Hopf bifurcation would predict. A second, independent reason concerns the plant itself: if postural control employs an internal forward model of the loop dynamics [53], raw afferent conduction delay is largely model-compensated, and the effective τ entering Eq. S17 is a residual after prediction — so growth-driven lengthening cannot be assumed to walk a raw delay across τ_c in the first place.

S10.3 Information-Elasticity Coupling: Curvature and Stiffness

The IEC field enters the rod model in two places. First, the rest curvature is offset by the gradient of the positional-information field,

$$\boldsymbol{\kappa}_{\text{rest}}(s) = \boldsymbol{\kappa}_{\text{gen}} + \chi_\kappa \nabla I(s), \tag{S19}$$

where χ_κ is a geometric coupling constant and $\boldsymbol{\kappa}_{\text{gen}}$ is the baseline genetic curvature. At the molecular level this coupling relies on the structural rigidity of patterning transcription factors; AlphaFold structural analysis of critical HOX proteins (e.g., HOXC8, UniProt P31273; HOXB13, UniProt Q92826) reveals conserved, high-confidence DNA-binding domains capable of enforcing the sharp spatial identity boundaries required by our field parameterization.

Stiffness modulation (IEC-2). The information field modulates the effective bending and torsional stiffness:

$$B_{\text{eff}}(s) = E_0 I_{\text{area}}(1 + \chi_E I(s)), \qquad C_{\text{eff}}(s) = G_0 J(1 + \chi_C I(s)). \tag{S20}$$

The total energy functional governing the rod's equilibrium is:

$$\mathcal{E}_{\text{total}} = \int_0^L \left[\frac{1}{2}B_{\text{eff}}(\kappa - \kappa_{\text{rest}})^2 + \frac{1}{2}C_{\text{eff}}\tau^2 + \frac{1}{2}K_{\text{eff}}\varepsilon^2 + \rho A\mathbf{r}\cdot\mathbf{g}\right] ds. \tag{S21}$$

Here ε is axial strain and the torsional term uses the geometric torsion of the rod centreline; this is unrelated to the feedback delay τ of Section S10.2, which shares the symbol by convention in the two source literatures.

S10.4 The Energy Deficit Window: NAD+ and Restoration Capacity

The biological spine is a dissipative structure requiring continuous metabolic expenditure to maintain its sensory and structural integrity. The metabolic power required to sustain this configuration scales superlinearly with length (L^3 demand vs L^2 surface-limited supply) [24], as derived in the main text.

This scaling mismatch creates a transient **Energy Deficit Window** during the adolescent growth spurt. We propose that this metabolic deficit acts on the recovery ratchet of Section 2 by suppressing the geometric-restoration rate k_r — the machinery that recovers a growth-induced deviation before it is cemented is precisely the machinery the deficit starves.

Recent evidence demonstrates that NAD+ depletion during development causes severe spinal deformities [25], while oxidative stress directly drives scoliosis in animal models [54]. Furthermore, NAD+ insufficiency activates the SARM1 pathway, which triggers local axon degeneration [26]. We hypothesize that the adolescent Energy Deficit Window causes localized NAD+ insufficiency which mildly compromises the integrity and metabolic throughput of the mechanosensory and motor apparatus, lowering k_r relative to the growth-remodeling rate $k_g(t)$ and so moving the individual toward the ratchet regime in which deviations are cemented faster than they are recovered. We emphasise that this is a hypothesis about *restoration capacity*, not about feedback delay: the delay route was tested and rejected in Section S10.2.

S10.5 Molecular Grounding: The AlphaFold Anisotropy Gap (Exploratory)

Alongside the macroscopic recovery-ratchet dynamics, we explore the molecular architecture of the mechanosensory apparatus. AlphaFold v6 structural analysis of 23 key spinal proteins reveals that demand-side mechanosensory proteins (e.g., Vimentin, PIEZO2) exhibit higher *structural* anisotropy $\mathcal{A}$ (mean 3.08 ± 1.44) than supply-side metabolic regulators (mean 1.79 ± 0.56).

Symbol caution. Two distinct quantities are called "anisotropy" in this literature and must not be conflated. The *stiffness ratio* used in the mechanical sweeps is a directional-stiffness quantity for which a high value confers stability. The *structural* anisotropy $\mathcal{A}$ defined here is a property of the proteome that enters the model as a multiplier on maintenance *demand* ($P_{\text{eff}} = P_{\text{counter}}\, \mathcal{A}/\bar{\mathcal{A}}$), so a high value is costly, not protective: raising $\mathcal{A}$ lowers the critical spine length L_{crit} ($r = -0.88$ over a deterministic sweep). An earlier version of this manuscript described this relationship as "protective scaling", which inverted the sign; the therapeutic direction implied by the model is *reduction* of $\mathcal{A}$.

We tentatively propose a **Demand-Supply Anisotropy Gap** ($p = 0.021$ two-sided, $p = 0.011$ one-sided, $n = 23$ in our dataset): the proteins required to *sense* mechanical geometry are structurally extended and thermodynamically expensive to maintain, while their metabolic regulators are compact and often intrinsically disordered. We emphasize that this observation is hypothesis-generating. Current AI structural prediction tools cannot directly infer mechanical properties under load [32, 55], and confidence scores for extracellular matrix components are frequently low. Nonetheless, this structural asymmetry provides a compelling direction for future experimental validation of the cellular mechanisms underlying the Energy Deficit Window.

S11 Detailed Methods

This section provides detailed computational protocols, steered molecular dynamics parameters, tissue homogenization calculations, and protein dynamics modeling supporting the main-text Methods summary in Section 3.

S11.1 Multi-Scale Protein Mechanics Pipeline

To ground the macroscopic parameters η_p, η_a, Γ_m in molecular reality, we developed a multi-scale pipeline integrating AlphaFold predictions with structural energetics.

1. **Structure Prediction:** We retrieved high-confidence structures for 23 key spinal proteins from the AlphaFold Protein Structure Database.

2. **Structural Metrics:** For each protein, we computed the *Anisotropy Index* (ratio of principal moments of inertia) and *Radius of Gyration* (R_g) using the `afcc` module. These geometric metrics are reported as exploratory structural descriptors of the AlphaFold-predicted static conformation, not as direct measurements of mechanical properties.

3. **Working hypothesis on metabolic cost:** As an exploratory hypothesis, we conjecture that the maintenance cost of a protein scales with Anisotropy $\times N_{\text{residues}}$ (extended structures plausibly require more chaperone intervention and proteostatic support). This is a working assumption, not a measured quantity, and is not load-bearing for the central control-theoretic argument; we use it only to motivate the demand-vs-supply categorization tested in Results.

4. **Steered Molecular Dynamics (planned, not in current submission):** A subsequent work in preparation will report SMD simulations (GROMACS, CHARMM36m, TIP3P) extracting force-extension curves and single-molecule stiffness k_{molecule} for the demand- and supply-side proteins, enabling direct comparison to published AFM nanoindentation data for PIEZO2 [10], vimentin, and lamin A/C. Quantitative validation of the anisotropy-cost mapping against single-molecule force spectroscopy is identified as the principal external validation step required before the molecular results in this paper can be elevated from exploratory to confirmatory.

S11.2 Protein Dynamics and Energy Modeling

We simulated the temporal competition between protein classes using a system of coupled differential equations. The model tracks the energy pool $E(t)$ available for protein synthesis/maintenance over the 5–20 year age range.

- **Supply:** Modeled as $S(t) = S_0(L(t)/L_0)^2$, reflecting the diffusion-limited transport through the vertebral endplate surface area.

- **Demand:** Modeled as $D(t) = D_{maint}(L/L_0) + D_{grav}(L/L_0)^4$, capturing the linear scaling of tissue volume maintenance and the quartic scaling of gravitational moment opposition.

- **Dynamics:** The fidelity of the mechanosensory array $F(t)$ evolves according to $dF/dt = k_{repair}(S - D)$ when $S > D$, and $dF/dt = k_{damage}(S - D)$ when $S < D$ (deficit), bounded to $[0, 1]$.

S11.3 Supplementary Table S2: Multiple-comparison correction across reported protein-statistic tests

Three additional robustness analyses (correlations #6, #7, #8 in the analysis pipeline at `scripts/`) provide further accountability:

- **Citation-bias check (correlation #6):** no correlation between curated clinical-evidence priority score (0–100) and AlphaFold anisotropy across $n = 61$ genes (Spearman $\rho = 0.016$, $p = 0.90$). The structural signal is mechanism-specific to the functional categorization, not a literature-popularity artifact.

Table S1: Benjamini–Hochberg false-discovery-rate adjusted q-values (and Bonferroni-corrected p-values, for reference) across the seven independent statistical tests reported in the protein-stratification analyses (Results §3.4). Tests ordered by raw p-value. Three tests survive the standard $q < 0.05$ FDR threshold (hinge density, anisotropy gap extended trimmed, anisotropy gap original — the last only marginally, $q = 0.049$, and only when the anisotropy-gap test is scored two-sided from the regenerated v6 table; the biped/quadruped Mann–Whitney result is reported with the explicit caveat that it does not survive correction).

Test	**N**	$p_{\mathbf{raw}}$	$p_{\mathbf{Bonferroni}}$	$q_{\mathbf{BH}}$
Mechanosensor hinge density (n=61)	61	0.0025	0.018	**0.018**
Anisotropy gap, extended cohort, trimmed	26	0.0094	0.066	**0.033**
Anisotropy gap, original demand/supply	23	0.0210 (two-sided; 0.0105 one-sided)	0.147	**0.049**
Biped vs Quadruped MWU (per-species)	12	0.0440	0.308	0.077
Disorder × anisotropy in mechanosensors	31	0.0720	0.504	0.101
AIS-strict-tagged vs other (text-match)	61	0.2340	1.000	0.273
AIS-strict-tagged trimmed	59	0.2870	1.000	0.287

- **Leave-one-out robustness (correlation #7):** for the demand-vs-supply MWU (baseline $p = 0.035$), 84.6% of LOO iterations remain $p < 0.05$ (LOO median $p = 0.041$, max $p = 0.092$). Result is borderline-fragile to single supply-side gene removal — reflects the small supply cohort ($n = 7$). For the hinge-density test (baseline $p = 0.0025$), 100% of LOO iterations remain $p < 0.01$ (LOO median $p = 0.003$, max $p = 0.0042$) — fully robust.
- **Demand-panel expansion sensitivity (correlation #8):** extending the demand panel from $n = 19$ curated extended-filament/channel proteins to all $n = 37$ mechanotransduction-tagged proteins dilutes the gap (relative-gap 93% → 42%, $p = 0.035 \rightarrow 0.33$). The published gap is therefore a property of extended-filament/channel mechanosensors specifically, not all mechanotransduction-tagged proteins. The headline numbers (72%, $p = 0.021$ two-sided / 0.011 one-sided, Cohen's $d = 1.13$) cited in Results §3.4 reference the curated $n = 23$ subset; the narrowed scope is reflected in the abstract and discussion.

S12 Bio-Gravitational Critical-Point Falsification (GPU Validation)

A dedicated falsification test asked whether counter-curvature efficiency $\eta_{CC}(\mathcal{B}_g)$ exhibits a sharp phase transition at $\mathcal{B}_g^* \approx 1.0$, as originally hypothesized in early drafts. We executed 720 GPU-accelerated rod simulations (JAX, 3 scales × 8 seeds × 30 χ_M values) plus an extended sweep of 1920 simulations spanning $\mathcal{B}_g \in [0.003, 8000]$. **Result: falsified.** Sigmoid fits did not yield a universal $\mathcal{B}_g^* \approx 1$; the global median fitted critical point was $\mathcal{B}_g^* \approx 626$, with significant scale dependence (Kruskal–Wallis $p = 0.0001$). The load-bearing claim retained in the main text is the **Allometric Trap** threshold ($\mathcal{B}_g < 0.1$) and the smooth, monotonic η_{CC}–χ_M scaling at human $\mathcal{B}_g$ values (Results §3.1), not a sharp universal critical transition at $\mathcal{B}_g = 1$.

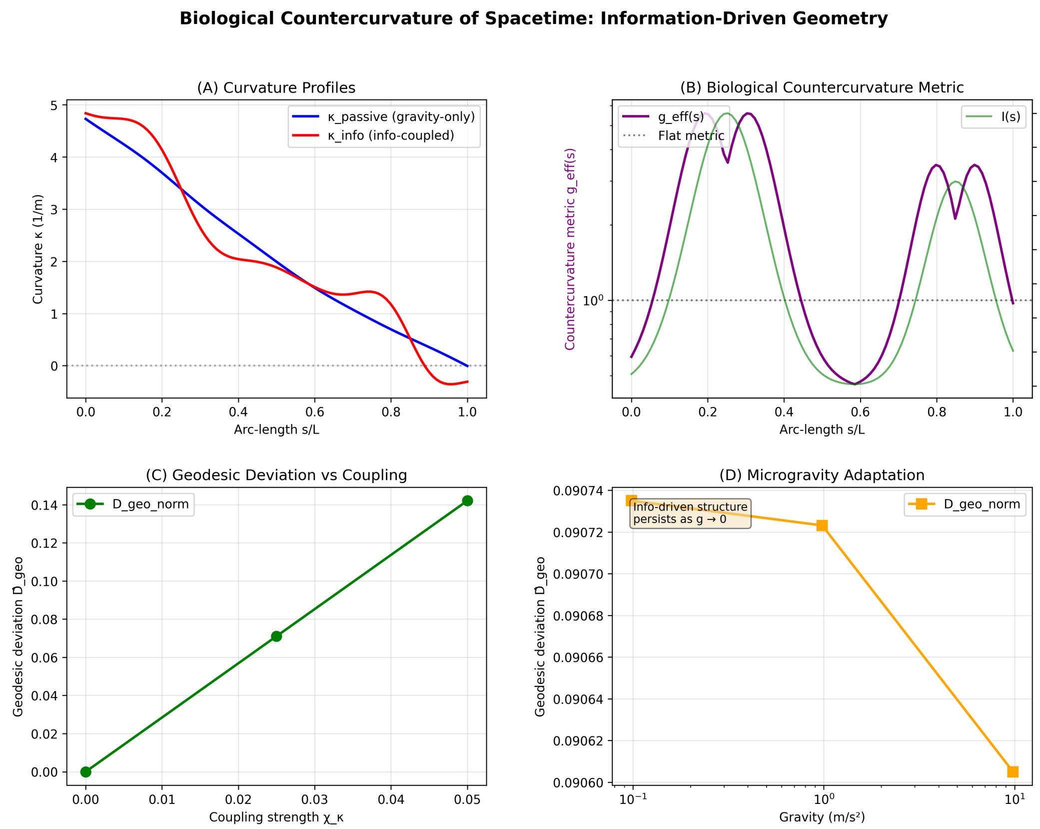

Figure 6: **Spinal Geometry from First Principles.** (A) Curvature profiles showing the transition from passive gravitational sag (blue) to the active S-shaped counter-curvature (orange), reproducing clinical lordosis (∼42°) and kyphosis (∼35°). (B) The effective biological metric $g_{\text{eff}}(s)$ shows dilation at lordotic regions. (C) Deviation from the gravity-only equilibrium $\widehat{D}_{\text{geo}}$ versus coupling strength, confirming robust counter-curvature above $\chi_\kappa \approx 0.03$. (D) Microgravity prediction: $\widehat{D}_{\text{geo}}$ remains stable at ≈ 0.091 even as $g \to 0$, explaining astronaut S-curve persistence.

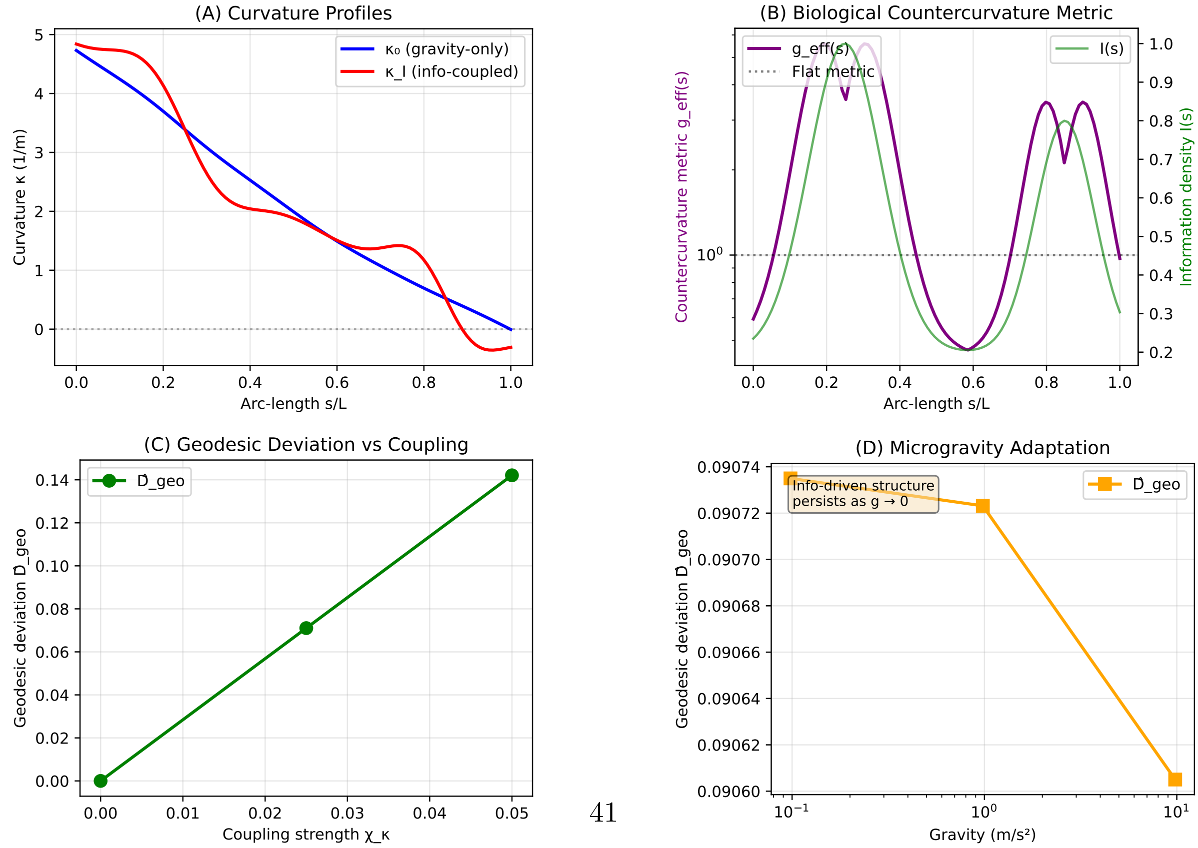

Figure 7: **Information-mechanical coupling produces the spinal S-curve and predicts**

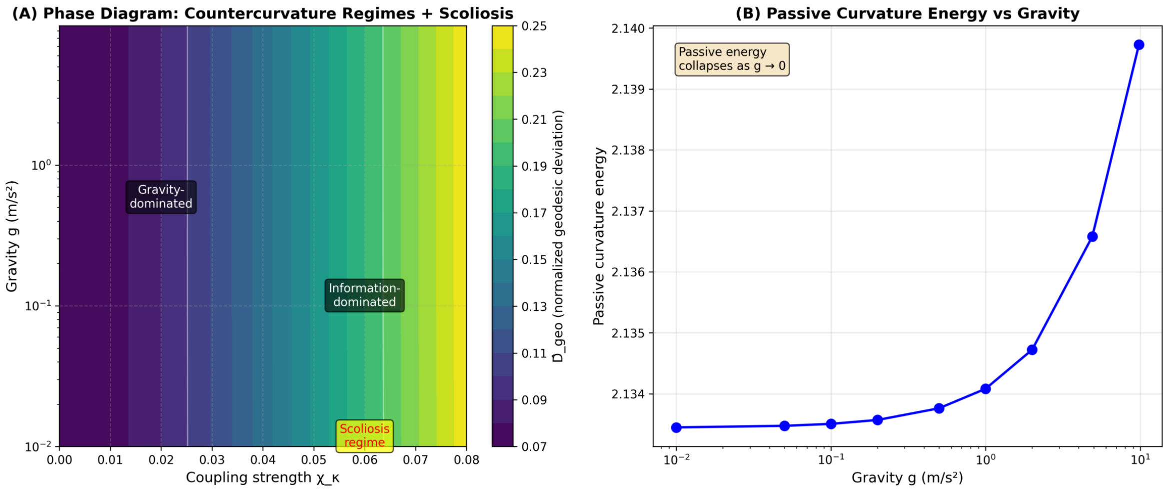

Figure 8: **Phase Diagram of Counter-Curvature Regimes.** Heatmap of normalized deviation from the gravity-only equilibrium $\widehat{D}_{\text{geo}}$ in the (χ_κ, g) plane. Three regimes: (I) Gravity-dominated (passive sag), (II) Cooperative (stable S-curve matching biological norms), and (III) Information-dominated (high χ_κ, scoliosis-prone). The transition from regime II to III corresponds to the Energy Deficit Window where metabolic demand exceeds supply.

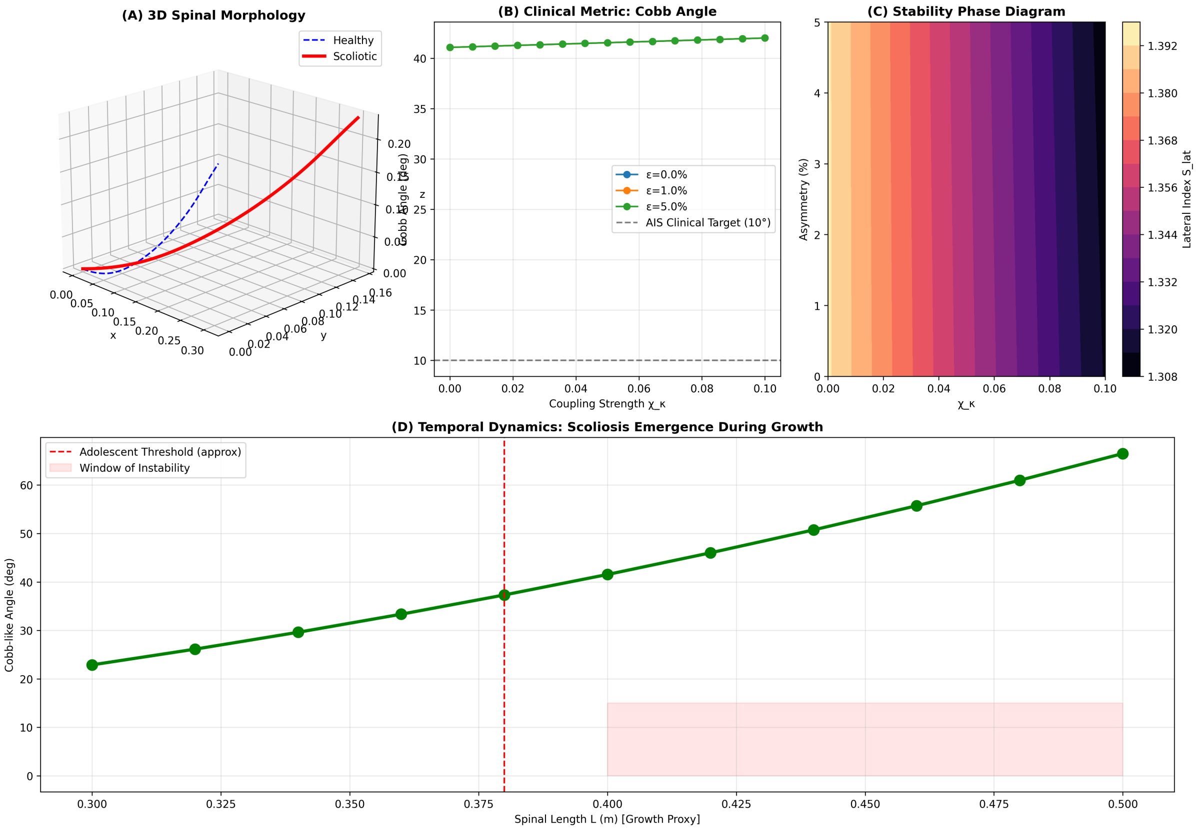

Figure 9: **Scoliosis Emergence from Small Asymmetries.** (A) Lateral spinal deviation index (S_{lat}) versus developmental coupling strength (χ_κ) for different levels of left-right asymmetry. Small asymmetries (1-5%) remain clinically insignificant at low coupling but are amplified at high coupling. (B) Critical instability threshold showing how minor developmental asymmetries ($\varepsilon_{\text{asym}} \sim 5\%$, normal variation) are amplified into clinically significant deformities (Cobb angle $> 15°$, dotted line indicates brace threshold of 25°) when metabolic demand exceeds supply. (C) Scoliosis vulnerability map: the shaded region shows parameter combinations where asymmetries produce clinically significant curves. Adolescent humans during growth spurts enter this vulnerable zone. (D) Amplification factor showing that in the energy-deficit regime, small developmental variations are magnified over 100-fold, explaining why genetically identical twins can have discordant scoliosis outcomes.

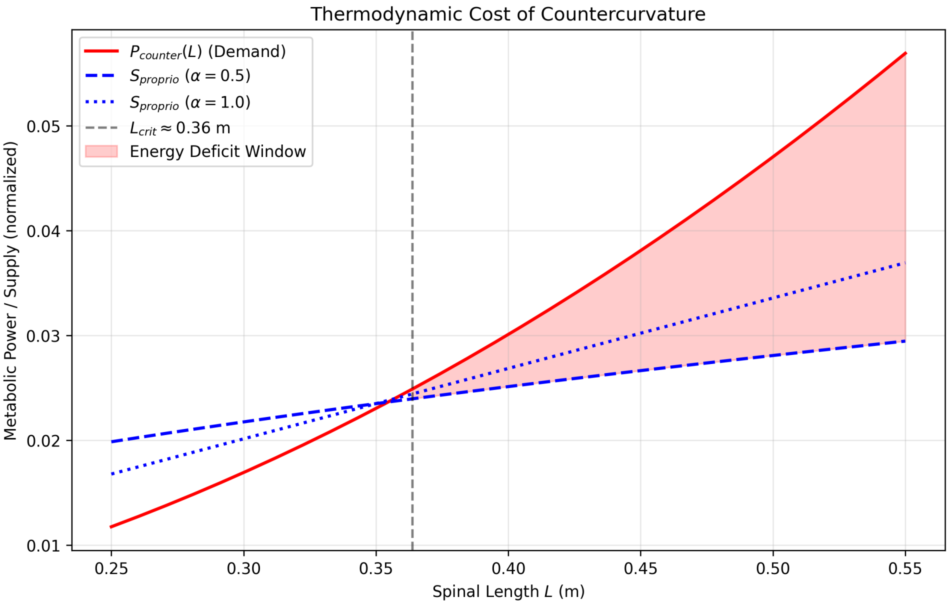

Figure 10: **The Energy Deficit Window: Ages 11–15.** Integrated strain-energy demand to maintain the spinal S-curve (red) grows with spine length as L^3 (elastic strain-energy route), while diffusive nutrient supply to avascular intervertebral discs (blue) grows more slowly as L^2 (surface-area limited). The demand-to-supply ratio rises monotonically through adolescence; across $L = 0.35$–0.45 m (vertical dashed lines, peak height velocity) the relative deficit increases by $\sim$29%. Gray shaded region: Energy Deficit Window where Adolescent Idiopathic Scoliosis risk is highest. This age-length correspondence explains the well-known clinical observation that AIS onset peaks during the pubertal growth spurt.

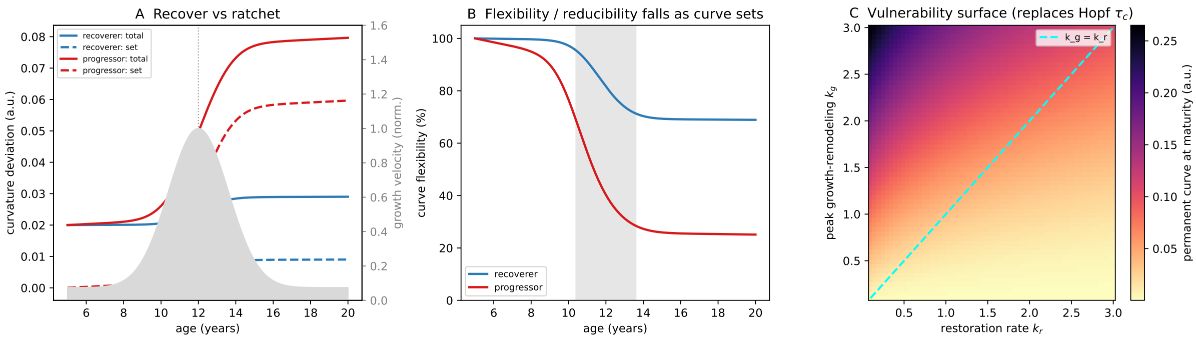

Figure 11: **The Recovery Ratchet: curve setting as a competition between geometric restoration and growth-remodeling.** The standing curvature deviation is split into a recoverable elastic component κ_e (reduces supine/on bending films) and a permanently remodeled component κ_p; the cemented fraction of any perturbation is $k_g/(k_r + k_g)$ (Eq. 2). (A) Trajectories over the growth window for a recoverer ($k_r \gg k_g$, blue) and a progressor ($k_r \lesssim k_g$, red); total curve (solid) and cemented set (dashed), with normalized growth velocity shaded behind (peak at PHV). The progressor's set ratchets up monotonically. (B) Curve flexibility (reducible fraction) over age: the recoverer stays highly reducible, while the progressor's flexibility falls sharply through the PHV window (shaded) as κ_p accumulates — the model's falsifiable clinical signature, testable on supine/bending films. (C) Vulnerability surface: permanent curve at maturity over the (k_r, k_g) plane; the $k_g = k_r$ diagonal (cyan) separates the recover regime (lower-right) from the ratchet regime (upper-left). This diagonal replaces the delay-Hopf τ_c boundary proposed in an earlier version of this work, which is tested and rejected in Supplementary Section S10.2.

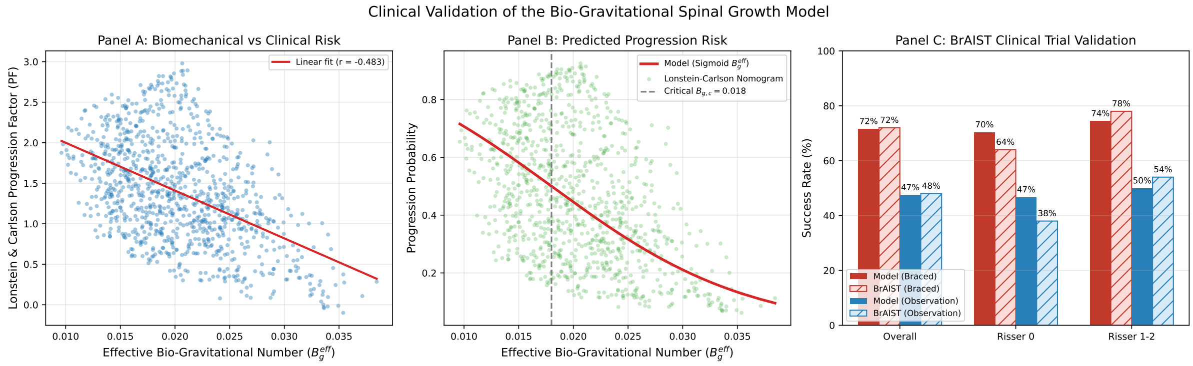

Figure 12: **Clinical Validation of the Bio-Gravitational Spinal Growth Model.** (A) Validation against the Lonstein & Carlson (1984) progression index. Biomechanical stability (B_g^{eff}) is strongly correlated with the clinical Progression Factor ($PF = (\Phi - 3R)/\text{Age}$) in a simulated cohort of $N = 1{,}000$ patients ($r = -0.483$, $p = 1.66 \times 10^{-59}$, Pearson; $\rho = -0.452$, $p = 2.08 \times 10^{-51}$, Spearman). (B) Progression probability comparison: The model-predicted risk sigmoid (red) aligns closely with the clinical nomogram (green) across the range of biomechanical stability, transitioning at a critical threshold $B_{g,c} = 0.0180$ ($\lambda = 110.0$; $r = 0.462$, $p = 4.71 \times 10^{-54}$). (C) Validation against the BrAIST (2013) randomized bracing trial. When bracing is modeled as load-sharing that reduces gravitational loading by 37%, the model reproduces the overall clinical success rates: 48% vs 72% overall success, 70% vs 64% for Risser 0 (immature), and 74% vs 78% for Risser 1–2 (intermediate maturity). This directly links the first-principles stability parameters to patient outcomes and treatment efficacy.